

Who Hauls the Habitat? A Verified Simulation Testbed, a Pooled Dispatch Marketplace and a Reference Design for Lunar Surface Logistics

Azizjon Rahimov

Roatify Research · Preprint, version 2, October 2026 · Code, scenario files, seeds, figure scripts and the browser testbed: lunar-logistics-research/ · Live simulator: [LunarSim Live](#)

Abstract. NASA's Moon Base plan lands about 4 tonnes of cargo at the lunar south pole before 2029, roughly 60 tonnes in 2029–2032 and 150 tonnes after that. Individual items range from 10 kg instruments to 15-tonne habitats, yet the rovers under contract carry 800–1,600 kg, and NASA's own architecture papers say that no planned vehicle can relocate a large element. NASA's April 2026 architecture guide names “a marketplace for lunar logistics” as a goal but does not say what one would do. We study that question with a discrete-event testbed of surface cargo movement at a nine-site south-pole base, verified against thirteen analytic and published references (all within 1% of their reference, the M/G/1 queueing check within 0.5%) and cross-checked against an independent browser implementation on 24 metric cells. The testbed models lander arrivals and manifests, a heterogeneous fleet owned by several companies, the lunar night, Earth–Moon control latency fitted to NASA's teleoperation trials, vehicle faults, vehicles' other duties and the return flow of trash and samples. We compare four dispatch rules: bilateral in-house fleets (today's contracting), pooled greedy dispatch, batched optimal assignment, and a pooled marketplace built on a sequential single-item auction with bundling, consolidation of bulk lots, team lifts for oversize items, night-aware acceptance and anticipatory staging before published landings. Across about 8,000 simulated years, with 30 common-random-number replications per cell and sequential replication to a 99% confidence interval within 1% of the mean for the headline cells, we find four things. First, pooling matters only when the market is fragmented: with one owner the marketplace and in-house dispatch are statistically indistinguishable, but with six owners sharing six vehicles the in-house 90th-percentile delay of urgent cargo is 44 h longer than the marketplace's; across 45 fleet configurations the in-house delay follows a single curve in the effective number of vehicles per owner, $25 + 20/e$ hours ($R^2 = 0.84$). Second, any pooled rule meets a 90% same-day service level for urgent cargo with four vehicles at Phase-2 demand where in-house fleets do not reach it with eight; because the fleet is 99% of the cost, a surface move costs about \$300 per kilogram of mass moved, or about \$5,000 per kilogram of landed cargo, against a landed cost of \$78,000–\$1,000,000 per kilogram. Third, the heavy-cargo gap is a property of the manifest, not of dispatch: a capacity ladder derived from NASA's cargo classes shows that 1.6-tonne rovers can move 40% of Phase-2 landed mass, two of them lifting together 55%, a 10-tonne hauler 88%, and only a 15-tonne capability reaches 100%. Fourth, teleoperation from Earth is workable at lunar latency with a staffed operator pool and collapses at Mars-like delays, where delivered mass falls below 10%. We also report what did not help: anticipatory staging, priority credits and backhaul changed nothing measurable, and we explain why. We close with a reference design for how a lunar logistics marketplace should be built and operated, and release the testbed, scenarios, seeds, scripts and a browser version that reproduces the paper's cells.

Keywords: lunar surface logistics, multi-robot task allocation, auction mechanisms, discrete-event simulation, verification and validation, horizontal collaboration, Artemis, Moon Base

1. Introduction

A decade of lunar planning has settled where cargo will land and how much of it there will be. NASA's *Moon Base Architecture User's Guide* (April 2026) describes three phases: 21 landings delivering about 4,000 kg to the

south pole in the first, 24 landings and about 60,000 kg in the second, 28 landings and about 150,000 kg in the third [1]. The agency's 2024 architecture white papers put recurring logistics at 2,000–6,000 kg per crewed mission, and occasional single elements such as habitats and pressurised rovers at 12,000–15,000 kg [2, 3]. Landing pads must sit more than a kilometre from the habitation zone because of plume ejecta, so everything that lands has to be moved, some of it as far as 5 km over slopes of up to 20 degrees [2].

Far less has been settled about who moves it. The two Lunar Terrain Vehicles (LTVs) ordered in May 2026, from Astrolab and Lunar Outpost at about \$220 million each, are crew rovers that carry 800 kg of cargo at full performance and roughly 1,600 kg at reduced performance [4, 5]. NASA's 2025 logistics workshop states the mismatch plainly: “no mobility assets exist that can relocate large elements” [5]. Each vehicle will belong to one company under one contract, and NASA has told the LTV providers they may sell rover time to other customers between NASA missions [6]. In January 2025 NASA paid nine companies \$24 million to study lunar logistics, including “surface cargo and mobility” and “business models” [7]. And on page 14 of the 2026 guide, the agency writes that “fostering a marketplace for lunar logistics will empower government and industry to develop the capabilities and operational competencies needed to support crewed missions to the Red Planet” [1].

That sentence is a goal, not a design. Nobody has said what a surface logistics marketplace would dispatch, with what rule, under which constraints, or whether it would help at all when the whole market is two rovers and a dozen cargo items a year. The question is not academic for us: Roatify is building a freight-dispatch platform for Earth trucking and intends to carry that software to the Moon in stages. Before raising money against that plan we wanted an honest answer to five questions.

1. **RQ1.** When does pooled dispatch beat bilateral in-house fleets, and by how much, for fleets of two to eight vehicles split among one to eight owners?
2. **RQ2.** How large is the heavy-cargo gap in delivered mass, how much of it can coordination (several rovers lifting one item together) recover, and what vehicle capacity closes it?
3. **RQ3.** How do autonomy level and control latency bound surface throughput, and how many Earth operators does a fleet need?
4. **RQ4.** What does a surface move cost per kilogram, how does it compare to the cost of landing that kilogram, and which assumptions drive it?
5. **RQ5.** How should a lunar logistics marketplace be built and operated: which functions, which rules, which data, and in what order?

Earth freight gives reasons for both optimism and caution. Pooling fleets across shippers saves 5–30% of transport cost in road and forest logistics [8, 9, 10]. But those gains come from thick markets. Theory on dynamic matching says that in thin markets greedy matching is close to optimal and waiting to batch requests mostly adds delay [11, 12]; empirical work on US trucking ties the efficiency of the spot market directly to its thickness [13]. The lunar market will be the thinnest freight market anyone has ever tried to run.

We answer the five questions with simulation, because there is no data to answer them otherwise, and we hold the simulation to a standard that makes its numbers worth quoting. We built a discrete-event testbed of cargo movement at a nine-site south-pole base and ran about 8,000 simulated years of operations. The testbed reproduces NASA's published spacing, slopes, cargo masses, lander cadence and rover capacities; it adds the lunar night, Earth–Moon control latency that interpolates NASA's own teleoperation trial table exactly, vehicle faults, vehicles' other duties and a return flow of trash and samples. Thirteen verification checks compare it with analytic formulas, published measurements and conservation identities; an independent implementation in

JavaScript reproduces its results within confidence intervals on every one of 24 metric cells; and the headline cells are replicated until the 99% confidence interval of the mean urgent delay is within 1% of the mean. On top of the testbed we implemented four dispatch rules, from today's bilateral contracting to a pooled marketplace that combines a sequential single-item auction with bundling, consolidation, team lifts, backhaul, night-aware acceptance and anticipatory staging. Our contributions are:

- We introduce **LunarSim**, an open, verified discrete-event testbed for lunar surface logistics with three demand phases, three vehicle classes, a sourced parameter table, fourteen experiment suites and a browser version that runs the same model interactively. Each parameter carries its source and a confidence grade, and each verification check its reference and tolerance.
- We propose a **pooled dispatch marketplace** for multi-owner rover fleets and give its clearing rule as a generalised-cost sequential auction (Algorithm 1), with team lift for items heavier than any single vehicle and anticipatory staging from the published landing schedule.
- We show **where pooling helps and where it does not**, and give the result a form that transfers: the gain is zero for a single owner and grows with fragmentation and other duties, and the in-house 90th-percentile delay across 45 fleet configurations collapses onto one curve in the effective number of vehicles per owner. Any pooled rule reaches a same-day service level with four vehicles at Phase-2 demand; in-house fleets do not reach it with eight.
- We quantify **three limits that no dispatch rule removes**, and give the first a general form: a capacity ladder that maps vehicle or team capacity to the share of NASA's manifest it can move; the collapse of teleoperation at Mars-like latency; and the fleet's fixed cost, which sets the price of a surface move regardless of how cleverly the fleet is dispatched.
- We give a **reference design** for how a lunar logistics marketplace should be built and served: its functions, data, rules, governance and a phased rollout tied to NASA's Moon Base phases.

Section 2 reviews what is known. Section 3 describes the testbed and its data. Section 4 defines the mechanisms. Section 5 gives the experimental protocol and the verification programme. Section 6 reports results, Section 7 presents the reference design, Section 8 discusses what the results mean for a marketplace operator and for NASA, and Sections 9–11 cover limitations, impact and conclusions. Appendices give the full parameter list, the verification table, the simulation reporting checklist and additional tables.

2. Background and related work

2.1 What NASA has published about surface cargo

Three NASA documents anchor our demand model. The *Lunar Mobility Drivers and Needs* white paper (2024) gives the range envelope (5–5,000 m), the ejecta standoff (>1,000 m), the slope envelope (up to 20°, with slopes over 10° “common”), the cargo classes (500–2,000 kg demonstrations, 2,000–6,000 kg recurring logistics per crewed mission, 12,000–15,000 kg habitation elements) and the capability expressed in the architecture (800 kg uncrewed on the LTV, about 1,500 kg planned in total) [2]. It lists the gaps the agency wants industry to fill: autonomy or tele-robotic operation through the whole year, standard cargo interfaces, offloading, levelling, mated power, and “the ability of multiple robotic mobility systems to work together” [2]. The *Lunar Surface Cargo* paper forecasts aggregate demand of 2,000–10,000 kg per year during the foundational phase, with occasional deliveries of up to 15,000 kg [3]. The 2026 architecture guide supplies the phase totals, a 10 km/h target for large rovers, a 2-tonne lander target for Phase 1, and the marketplace sentence quoted above [1].

2.2 Space logistics modelling

Campaign-level space logistics has a mature modelling literature. MIT's SpaceNet and the time-expanded network models of Ho, de Weck and colleagues optimise commodity flows between Earth, orbit and the lunar surface over mission campaigns [14, 15, 16]. These models decide how much mass to send and when; they treat the surface as a single node. The surface itself has been studied for traverse planning and rover energy budgeting (VIPER's planning deck gives speeds of 10–20 cm/s while driving and a mission average of 1 cm/s [17]), and for cooperative transport of large payloads at the control level [18, 19]. We found no model of dispatching cargo among several companies' vehicles on the surface, and no treatment of lander-arrival lumpiness, lighting windows and latency together.

2.3 Multi-robot task allocation and market-based coordination

Our mechanism sits in the market-based multi-robot task allocation (MRTA) tradition surveyed by Dias et al. [20] and formalised by Gerkey and Mataric [21]. The sequential single-item (SSI) auction of Koenig, Lagoudakis and colleagues allocates tasks one at a time to the robot whose route cost increases least; its total path cost is within a factor of two of optimal under symmetric, triangle-inequality costs, and it needs at most $|\text{tasks}| \times |\text{robots}|$ bids [22, 23]. The consensus-based bundle algorithm (CBBA) decentralises the same idea [24]. These methods assume one cooperative owner and have no notion of price or of items that need two robots. We keep the SSI structure, replace route cost with a generalised cost that includes the shipper's value of time, and add a team-lift stage.

2.4 Freight marketplaces, collaboration and market thickness

Combinatorial auctions for truckload procurement [25], horizontal collaboration among carriers [8, 9] and digital freight matching [26] all report that pooling demand across shippers cuts cost, typically by 5–30%, and that the hard part is a fair split of the gain [9, 10]. Dynamic matching theory adds the thin-market caveat: without information about which requests are about to expire, greedy matching is near-optimal and batching mostly adds waiting [11]; a windowed matching that waits $d+1$ periods is only 0.279-competitive [12]. On US spot trucking, Harris and Nguyen find efficiency rising with thickness [13]. Even in that market only about 15% of loads clear on the spot exchange; the rest move under contracts [27]. In dynamic vehicle routing, exploiting known future demand to reposition vehicles is a classic lever [58]; the lunar version of that lever is the published landing schedule, which we test in Section 6.11.

2.5 Teleoperation under delay and supervisory control

Round-trip light time to the Moon is about 2.6 s; with relays and ground processing NASA quotes up to 10 s one way [28]. NASA JSC's LTV teleoperation trials measured average driving speeds of 3.24, 2.56, 2.03 and 1.76 km/h at 0, 4, 6 and 8 s of round-trip delay [29]. A lunar-excavation teleoperation study found task completion times of 139% and 174% of baseline at 1.5 s and 3 s delay, with success rates falling from 91% to 45% [30]. Operators adopt a move-and-wait strategy once delay exceeds about a second [31]. For supervisory control, Olsen and Wood's fan-out metric gives 1–9 robots per operator depending on autonomy [32], and mining autonomous haulage runs up to about 30 trucks per controller [33]. We interpolate the JSC table exactly in Section 3.4 and sweep operator fan-out in Section 6.5.

2.6 Simulation verification and reporting

Our verification programme follows the standard texts: analytic queueing references for the engine core, common random numbers with paired confidence intervals, and sequential replication until a target precision is reached [43, 59]. Reporting follows the ODD protocol for agent-based models [34] and the STRESS guidelines for discrete-event simulation [35]; the cross-implementation check is in the spirit of the reproducibility checklists now common in machine-learning venues [57].

Table 1. Where this work sits relative to the nearest prior families. A tick means the family models the feature; a dash means it does not.

Family	Multi-owner fleet	Prices / bids	Oversize items (team lift)	Lunar night	Control latency	Lander lumpiness	Verified, open code
Campaign logistics (SpaceNet, time-expanded networks) [14–16]	–	–	–	partly	–	✓	partly
MRTA auctions (SSI, CBBA) [22–24]	–	bids, no payments	–	–	–	–	partly
Freight matching and collaboration [8, 9, 25, 26]	✓	✓	–	–	–	–	–
Teleoperation studies [29–32]	–	–	–	–	✓	–	–
This paper (LunarSim)	✓	✓	✓	✓	✓	✓	✓

3. The testbed

LunarSim is a discrete-event simulation written in Python 3.12 with NumPy; SciPy is optional (used for the assignment baseline when present, with a built-in Hungarian implementation otherwise). Time is measured in hours; a run covers one Earth year of operations. A line-by-line port to JavaScript runs the same model in a browser page and is used both as an interactive tool and as an independent implementation for cross-checking (Section 5.4). This section is a summary ODD description [34]; the full parameter list with sources is in Appendix A, the verification table in Appendix B and the STRESS reporting checklist [35] in Appendix C.

3.1 Purpose and entities

The model exists to compare dispatch rules for a shared pool of rovers that move landed cargo at a south-pole base, under the constraints NASA has published. Three kinds of entity flow through it. *Cargo items* have a mass, an origin pad, a destination site, a release time (when offloading finishes), a priority class (1 urgent, 2 normal, 3 deferrable), an owner, an interface standard and a class label. *Vehicles* have an owner, a class (capacity, speed, energy, battery, night capability, team-lift capability, unit cost, design life, fault parameters, handling times), a location, an autonomy level and a set of accepted interface standards. *Jobs* are what a dispatch rule returns: one or more vehicles, a bundle of items sharing an origin, a visiting order of destinations, and optional backhaul items. A fourth entity, the *landing schedule*, is known in advance: real landings are published months ahead, and the anticipatory component of the marketplace reads it. Figure 2 shows how the pieces connect.

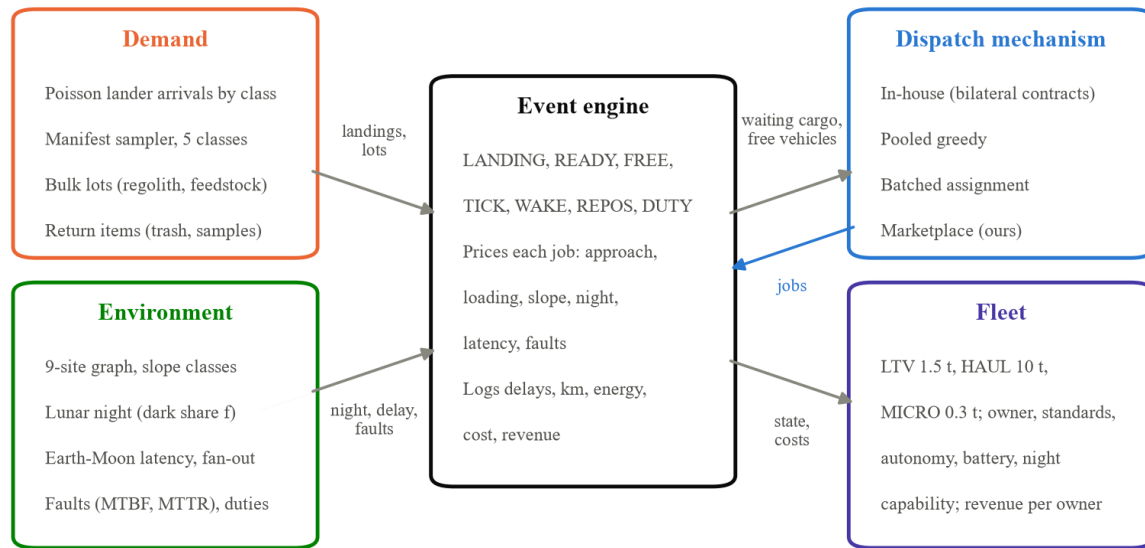


Figure 2. Testbed architecture. The demand and environment submodels feed an event engine that calls a dispatch mechanism whenever cargo becomes ready, a vehicle becomes free, a night ends or a landing is near (or at fixed ticks for batched mechanisms). The engine prices every returned job: empty approach, loading, slope- and night-adjusted travel, teleoperation slowdown, operator availability, random faults and a hibernation penalty if a vehicle without night capability is caught in the dark.

3.2 Sites and terrain

The base is a nine-site graph on a local tangent plane (Figure 1). Two landing pads sit 1.7 and 2.1 km from a cargo depot beside the habitation zone, satisfying the >1 km ejecta standoff. A power site, a construction and excavation site and an in-situ resource (ISRU) pilot plant lie 0.6–2 km from the base; a science site and a shadowed crater rim (PSR) lie about 3 and 5 km out, matching NASA's 5,000 m aggregation range. Each edge carries a slope class (flat, 0–5°; moderate, 5–10°; steep, 10–20°) that multiplies travel time by 1.0, 1.33 or 2.0 and energy by 1.0, 1.35 or 1.9. Straight-line distances are scaled by a tortuosity factor of 1.3 to represent paths that wind around craters and boulders. Vehicles take shortest-time paths, computed once by Floyd–Warshall.

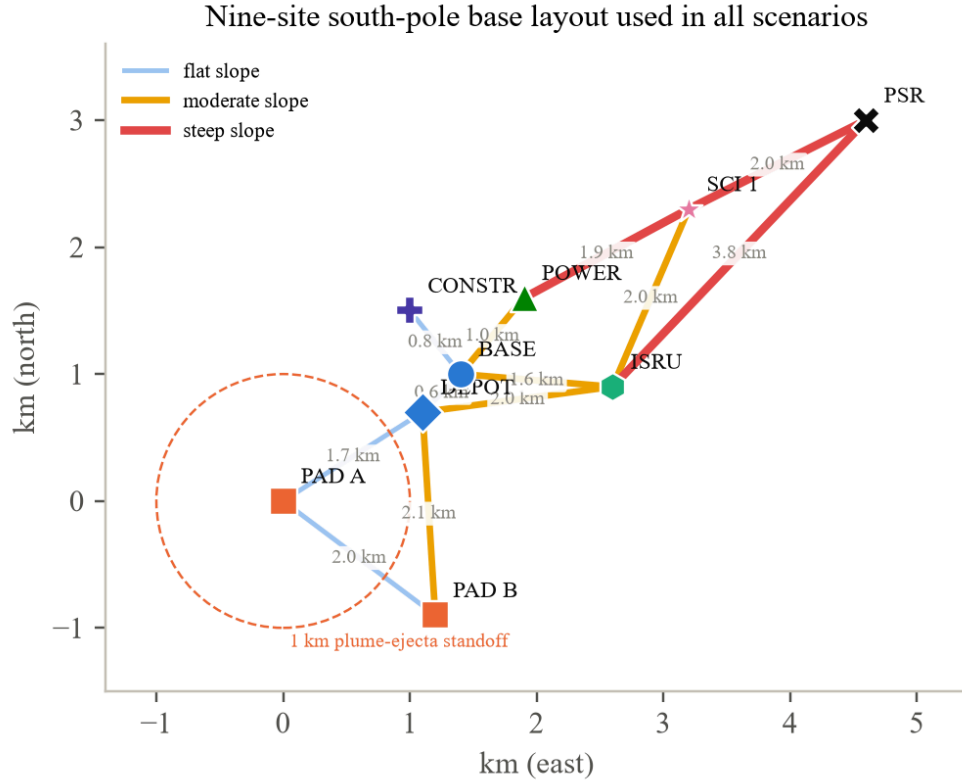


Figure 1. The nine-site south-pole base used in every scenario. Edge labels give path lengths after the tortuosity factor; colours give slope class. The dashed circle marks the 1 km plume-ejecta standoff that keeps pads away from the base.

3.3 Demand

Landers arrive as independent Poisson processes per class, each landing at a pad drawn when the schedule is generated. CLPS-class landers deliver 100–500 kg (Firefly Blue Ghost, Intuitive Machines Nova-C, Astrobotic Griffin); MK1-class landers deliver 2,000–3,000 kg (Blue Moon MK1); large landers deliver 12,000–30,000 kg (Starship HLS cargo, Blue Moon MK2). Each landing draws a manifest from five cargo classes (Table 2) until the payload mass is used up, with the lander's crane limiting the heaviest single item. Items become available 6–72 h after touchdown depending on lander class, staggered by 15 minutes per item. Three phases scale the cadence to match NASA's totals: Phase 1 (about 5 t/yr of landed cargo, 19 items), Phase 2 (about 30 t/yr, 49 items) and Phase 3 (about 74 t/yr, 79 items). Figure 3 shows the manifest-based landed-mass forecast we assembled from public mission plans and the simulated item-mass distribution by phase.

Two further flows matter for a dispatch rule and are usually omitted. *Bulk lots* represent regolith moved for berms and shielding and excavated feedstock carried to the ISRU plant; they are released evenly through the year in 500 or 1,000 kg lots as deferrable (priority 3) items. NASA sizes its IPEX excavator demonstration at 10,000 kg of regolith per day [36], so our Phase-2 default of 500 t/yr and Phase-3 default of 1,500 t/yr are deliberately modest; Section 6.7 sweeps 0–3,000 t/yr. *Return items* represent trash, samples and empty carriers: 30% of outbound items generate a return item of 20–60% of their mass, ready 5–30 days after delivery, bound for a pad. Both are drawn from the demand random stream so that every mechanism sees the same flows.

Table 2. Cargo classes used by the manifest sampler. Masses are uniform within the range; destinations are drawn with the listed weights. Shares are per item for a build-out manifest; the lander's payload and crane limit truncate them.

Class	Mass (kg)	Destinations (weight)	Priority	Item share	Anchor
Consumables and spares	60–400	BASE 0.8, DEPOT 0.2	1	0.34	Logistics carriers; packaging overhead 75–150% [37]
Science payloads	10–150	SCI 0.5, PSR 0.3, BASE 0.2	2	0.20	CLPS payload masses 1–100 kg [38]
Equipment and demos	300–2,000	CONSTR 0.4, ISRU 0.3, POWER 0.2, BASE 0.1	2	0.26	“500-to-2,000 kg” demonstrations [2]
Infrastructure	2,000–6,000	POWER 0.4, ISRU 0.3, CONSTR 0.3	2	0.14	Logistics 2,000–6,000 kg per mission [2]
Elements	6,000–15,000	BASE 0.6, POWER 0.4	3	0.06	Habitats, power units 12,000–15,000 kg [2, 3]
Bulk lots (regolith, feedstock)	500 or 1,000	CONSTR to BASE / ISRU; PSR to ISRU	3	scenario	IPEX 10 t/day [36]
Return items	20–60% of parent	back to the origin pad	3	30% of items	Trash management, sample return [1, 7]

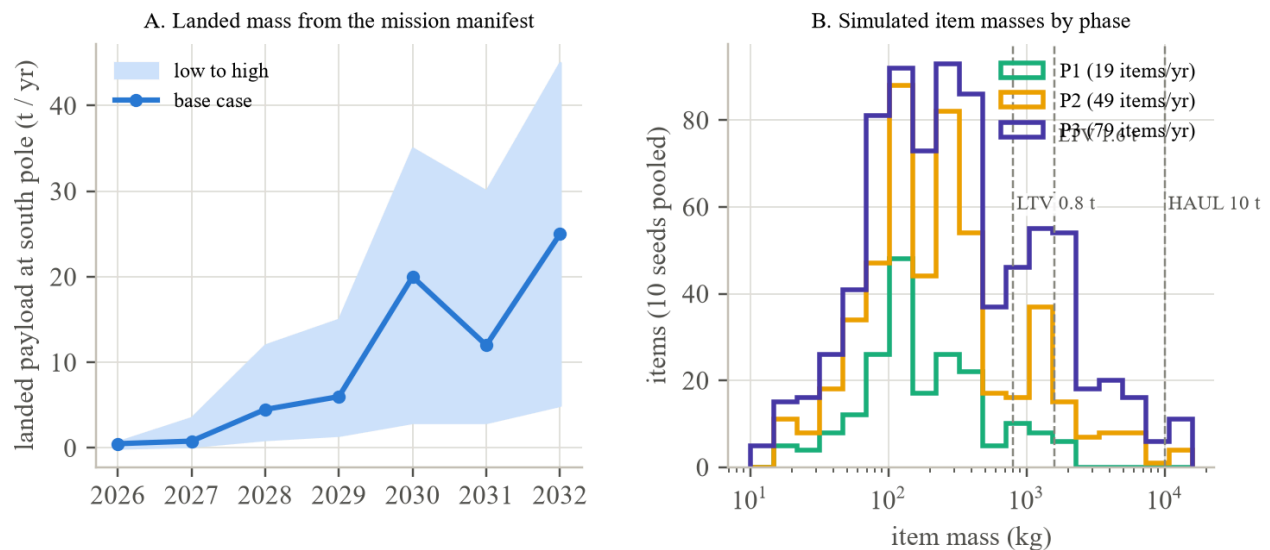


Figure 3. Demand. **A:** landed payload mass at the south pole by year, assembled from public mission manifests (low, base and high cases; Appendix A lists each mission). The 2026–2028 base case is five or six discrete sub-3-tonne items, which is the single most important fact for a marketplace design. **B:** item-mass histograms from ten simulated years per phase. Vertical lines mark the LTV's 800 kg full-performance and 1,600 kg reduced-performance cargo limits and the 10-tonne hauler we introduce.

3.4 Environment submodels

Lunar night. A synodic month is 708.7 h. A fraction f of each cycle is dark at the base; the default is 0.15 (about 106 h), consistent with 90–94% illumination on the Shackleton rim [39] and NASA's 120-hour darkness survival target [1]. Vehicles without night capability cannot start jobs in the dark; if one is caught out when night begins it hibernates until dawn, and the job is delayed by the whole night. Night-capable vehicles drive at 60% speed in the dark.

Control latency and autonomy. Each vehicle is teleoperated, supervised or autonomous. Round-trip control latency is 2.6 s of light time plus a ground-processing term, 2.4 s by default (RTT 5 s). Teleoperated driving speed interpolates the JSC trial table exactly (3.24, 2.56, 2.03 and 1.76 km/h at 0, 4, 6 and 8 s [29]); beyond 8 s it follows the move-and-wait law $v = v_g(1 + 8k)/(1 + k \cdot \text{RTT})$ with $k = 0.1 \text{ s}^{-1}$ (a 10 s command segment), which gives a few tens of metres per hour at Mars delays, in line with command-cycle driving on Mars rovers. Speeds are relative to the 6 km/h autonomous average we assume for a 10 km/h-class rover. Teleoperated loading and unloading take $(1 + 0.25 \cdot \text{RTT})$ times nominal, fitted to the excavation study [30]. Supervised vehicles drive at 0.9 of autonomous speed with a 1.1 handling factor. Earth operators are a shared resource: a teleoperated vehicle needs one operator for the whole job, a supervised vehicle 1/4 of one (fan-out 4) and an autonomous vehicle 1/20. A job that cannot get its operator waits.

Faults and other duties. Each vehicle fails after an exponentially distributed number of operating hours (mean MTBF 300–500 h by class) and recovers after a fixed MTTR of 12–36 h; this yields 92–96% availability, in the range reported for autonomous mining haulage [40]. Vehicles also spend a share of their time (default 30%) on their owner's other work in 48-hour blocks, reflecting NASA's statement that LTVs serve NASA missions first and may be hired out in between [6].

Energy. Driving energy is 350 Wh/km for an LTV-class rover (an Apollo LRV-derived design value is about 95 Wh/km for a 700 kg vehicle [41]; we scale up for mass and slopes), rising with payload and slope class. A vehicle whose battery cannot cover a job charges first at 5 kW. Energy never binds in the scenarios reported here, so we do not discuss it further.

3.5 Vehicles and costs

Three classes (Table 3) span the fleet that is plausible before 2033. The LTV class matches the contracted rovers: 1,500 kg cargo, 10 km/h design speed modelled as a 6 km/h autonomous average, survival but not operation through the night. The 10-tonne hauler does not exist and is our proxy for the “high capacity mobility systems” NASA lists as a partnership priority [1]; its cost is an assumption built from a \$200 million vehicle plus 4 t landed at Starship's published \$100,000/kg. The small utility rover matches Lunar Outpost's HL-MAPP class. Delivered cost per vehicle combines the vehicle contract with its landing (the Phase-1 LTV task orders were about \$220 million each plus about \$234 million per Blue Moon MK1 delivery [4]).

Two cost views appear in the results. *System cost* is the fleet's amortised capital (unit cost over calendar life) plus operator hours at a loaded \$1,000 per console hour; divided by delivered mass it gives cost per kilogram moved, and divided by landed cargo alone it gives cost per landed kilogram. *Variable job cost*, used inside the auction, is the usage-driven share (50%) of a vehicle's per-operating-hour cost plus operator time. Both are assumptions and both are swept in Section 6.10.

Table 3. Vehicle classes. Sources and confidence grades for each entry are in Appendix A.

Class	Cargo (kg)	Speed, autonomous (km/h)	Night ops	Team lift	Delivered cost (\$M)	Life (yr)	MTBF / MTTR (h)	Load / unload (h)
LTV (Astrolab, Lunar Outpost class)	1,500	6.0	no	yes	450	10	500 / 24	0.5 / 0.5
HAUL (proposed heavy hauler)	10,000	4.0	yes	yes	600	10	400 / 36	1.5 / 1.5
MICRO (HL-MAPP class)	300	3.0	no	no	60	5	300 / 12	0.3 / 0.3

3.6 Scheduling and outputs

The engine keeps an event heap ordered by time and insertion sequence. Landing events create manifests and schedule READY events; READY, FREE (a vehicle finishes), NIGHT_END, WAKE (two reminders before each published landing, and a timer when bulk lots are held) and REPOS (a scheduled staging move) events trigger a dispatch call for event-driven mechanisms; batched mechanisms act only at TICK events. When a mechanism returns jobs, the engine computes each job's duration from the slowest vehicle's approach, team set-up (1 h), handling, loaded legs, backhaul legs and charging, then adds fault recovery and night hibernation where they apply, and books operators. Delivery times for intermediate stops are interpolated along the job. The engine logs, per run, delivered and stranded mass, delays by priority and class, trips, bundled trips, team lifts, backhaul mass, repositioning moves, kilometres and empty kilometres, energy, utilisation, operator hours, faults, night strandings, costs and revenue per owner; with tracing on, it also records every vehicle movement for replay. An item is *stranded* if no vehicle in the fleet, or no feasible team of up to three team-capable vehicles when the mechanism supports teams, can ever carry it.

4. Dispatch mechanisms

All four mechanisms see the same information at dispatch time: the waiting items that are ready and not stranded, the vehicles that are free, alive, not on other duty and able to operate in the current lighting, and the published landing schedule. None sees future cargo masses.

In-house fleets (status quo). Every cargo owner is served by one contracted carrier: itself if it owns vehicles, otherwise a carrier assigned round-robin at the start. The carrier serves its customers' oldest item first with its nearest feasible vehicle, one item per trip. This is bilateral contracting as practised in CLPS and the LTV task orders.

Pooled greedy. Identical to in-house except that every vehicle may serve every item. This is the “closest driver” rule of ride-hailing and the simplest thing a marketplace could do.

Batched assignment. Every hour, solve a minimum-cost assignment (Hungarian algorithm [42]) between free vehicles and waiting items, with cost equal to the single-item job duration minus a small age credit so that old items are not starved. One item per vehicle. This is Uber-style batched matching and is optimal within each batch.

Marketplace (ours). Algorithm 1 gives the clearing rule. It differs from pooled greedy in seven ways, each of which Section 6.8 removes in turn. (i) *Team lift*. Items heavier than any free vehicle are assigned first to the

smallest team of two or three team-capable vehicles whose capacities sum to the item's mass; the team couples at the pad (1 h), drives at half speed and shares the job. (ii) *Generalised-cost bids*. For every (vehicle, item) pair the vehicle bids its variable cost for the job plus the shippers' value of the delivery time, minus an urgency credit for priority and waiting age; the globally lowest bid wins, and the auction repeats on the remaining pool (the SSI structure of [22]). (iii) *Bundling*. Before bidding, a vehicle greedily adds waiting items from the same pad while capacity allows and the detour costs less than 80% of a separate trip would; up to four distinct destinations are visited in shortest order. (iv) *Consolidation*. Deferrable bulk lots on a lane are held back until they would fill 70% of the best available vehicle or the oldest has waited 24 h, whichever comes first; a wake-up event guarantees release. (v) *Backhaul*. A vehicle finishing at a site loads any waiting return items there that fit and are bound for a pad. (vi) *Night awareness*. A vehicle without night capability refuses a job it could not finish before dark. (vii) *Anticipatory staging*. When a landing is due within 48 h and no vehicle is at or heading to its pad, an idle vehicle that the auction did not need is sent so that it arrives one hour before offloading can finish; staging never pre-empts waiting cargo and never sends a vehicle that would be caught by the night.

$$\text{bid}(v, B) = \sum_{u \in \text{team}} h(v, B) \cdot [\omega \cdot c_u + o_u \cdot p_{\text{op}}] + \sum_{i \in B} \text{VOT}(q_i) \cdot h_i(v, B) - \sum_{i \in B} [w_{q(i)} + w_{\text{age}} \cdot (t - r_i)] \cdot u \quad (1)$$

Here $h(v, B)$ is the job duration for vehicle v carrying bundle B from the current state, c_u the per-operating-hour cost of vehicle u , $\omega = 0.5$ the usage-driven wear share, o_u the operator demand (1, 1/4 or 1/20), $p_{\text{op}} = \$1,000/\text{h}$, $\text{VOT}(q)$ the value of delivery time by priority class (\$5,000, \$1,000 and \$100 per hour), h_i the time at which item i is dropped, r_i its release time, $w_1 = 6$ h, $w_2 = 2$ h, $w_3 = 0$, $w_{\text{age}} = 0.08$ h per hour waited, and $u = \$2,000$ per hour of credit. Section 6.9 searches over w_1 , w_2 , w_{age} and the consolidation thresholds.

Algorithm 1. Marketplace clearing (called at every READY, FREE, NIGHT_END, WAKE or REPOS event)

1. **Input:** waiting items P , free vehicles F , time t , published landing schedule L .
2. **Consolidation:** for each lane of deferrable bulk lots whose total mass is below 70% of the best feasible vehicle's capacity and whose oldest lot is younger than 24 h, remove the lots from P and schedule a WAKE event at the 24 h mark.
3. **Team lift:** for each item in P heavier than every vehicle in F , in priority-then-age order, find the smallest set of 2–3 free team-capable vehicles with enough combined capacity; if the team can finish before dark (or works at night), create the job and mark the vehicles used.
4. If P has more than 40 items, keep the 40 most urgent and oldest for this call.
5. **Auction:** while unused vehicles and items remain: for every feasible (vehicle, item) pair, build a bundle by insertion and compute bid (1); take the lowest bid. If the winning vehicle cannot finish before dark, idle it and continue. Otherwise append any return items at the last stop that fit (backhaul), create the job, remove its items from P .
6. **Anticipation:** for each landing in L within 48 h whose pad has no vehicle present or en route, pick the nearest unused free vehicle, and if it can reach the pad before dark, schedule a REPOS event so that it arrives one hour before the first item can be ready.
7. **Output:** the list of jobs. The engine prices and executes them.

Every job is paid at its bid-side variable cost, so revenue by owner is a by-product of the run; we use it only to report how evenly a pooled market spreads work across owners (Gini coefficient of revenue). A full treatment of payments, gain sharing and incentive compatibility is outside this paper; Section 7 places it in the reference design.

5. Experimental setup, verification and validation

Scenarios. Table 4 lists the fourteen experiment suites. Each is built from the three phases by changing one or two factors. Phase 2 (2029–2032, about 30 t/yr landed plus 500 t/yr of bulk lots) is the central case because it is the first period with enough cargo and enough vehicles for dispatch to matter and the period NASA has earmarked for logistics build-out.

Replications and variance reduction. Every cell is run with 30 seeds (20 in the latency suite, which has 54 cells). The demand stream (landings, pads, manifests, bulk lots, return items) and the operations stream (faults, duty blocks) are separate generators seeded from the cell seed, so for a given seed every mechanism faces the same cargo and the same fault history: common random numbers. We report means with 95% confidence intervals (1.96 standard errors) across seeds and, for mechanism comparisons, paired differences with their own intervals. A one-year horizon starts from an empty base; because landers arrive within days and the backlog never accumulates across lunar cycles in the reported cells, we do not discard a warm-up period.

Metrics. The primary metric is the 90th-percentile delay of urgent (priority 1) items from release to delivery, with the share of urgent items delivered within 24 h as its service-level form. Secondary metrics: delivered share of mass, immovable share of landed cargo mass, delay by cargo class, trips per year, bundled trips, team lifts, repositioning moves, backhaul mass, empty-kilometre share, utilisation, operator hours, faults, night strandings, system cost per kilogram and revenue Gini. Delay of deferrable bulk lots is reported separately because the marketplace delays them on purpose.

Tuning. The marketplace's hand-set weights were chosen once from a three-seed pilot and then frozen. Section 6.9 reports a random search of 40 weight configurations on Phase 2 with eight seeds and tests the winner on all phases with 30 fresh seeds. The baselines have one tunable each (the batching window and the age credit of batched assignment); we report the batching sweep in Section 6.8 and fixed the age credit at its pilot value.

5.1 Verification against independent references

Thirteen checks compare the simulator with something it did not produce (Appendix B, Table B1). Four are kinematic and structural identities: travel time on a flat path equals distance over speed; a steep edge doubles travel time; the dark share of simulated time equals the night parameter; the annual fleet charge equals the sum of unit costs over calendar lives. Four reproduce the measured teleoperation speeds at 0, 4, 6 and 8 s round-trip delay [29] to within 1%. Two are conservation identities under two different mechanisms: ready mass equals delivered plus stranded plus backlog mass to machine precision. One checks the stranding rule: in six large landings with a two-LTV fleet, no item heavier than the 3,000 kg team limit is ever delivered. One checks common random numbers: four mechanisms run on the same seed see an identical landed manifest. The last and most demanding check is a queueing reference: with one autonomous rover, no night, no faults, Poisson single-item arrivals and a fixed destination, the simulated mean wait in queue must match the Pollaczek–Khinchine M/G/1 formula $W_q = \lambda E[S^2]/(2(1-\rho))$. Over 18,000 items it agrees to 0.5%. All thirteen pass, with a maximum relative error of 0.5% and all but one at or below 0.03%. The same checks are implemented in the browser version and can be run by the reader.

5.2 Validation against published behaviour

Where no formula exists we compare behaviour with published anchors. Pooling gains in collaborative road and forest logistics are 5–30% of cost [8–10]; our pooled-versus-in-house gains in urgent delay are larger than that band when fleets are fragmented and zero when they are not, which the thickness literature predicts [11, 13].

Autonomous haulage availability of 88–95% [40] brackets our 92–96%. Earth truckload deadhead is 16–21% [27]; ours is about 50% because almost all lunar cargo flows one way, a structural difference we expected and discuss in Section 6.1. The lunar night's effect, a whole-night loss whenever a non-night-capable vehicle is caught out, matches the operational logic in NASA's survival targets [1].

5.3 Statistical precision: replicating until the estimate is settled

Thirty replications give confidence intervals of about $\pm 15\%$ on mean urgent delay and $\pm 40\%$ on its 90th percentile for a single cell, which is enough to rank mechanisms with paired differences but not to quote a single number to a decimal. For the four Phase-2 default-fleet cells we therefore ran the sequential procedure of Hoad, Robinson and Davies [59]: add replications in blocks of 100 until the 99% confidence interval of the mean urgent delay, the same-day service level, the delivered share and the cost per kilogram are each within 1% of their means, and the 90th-percentile delay within 5%, up to a cap of 4,000 replications. Section 6.14 reports the replications needed and the settled estimates; the headline Phase-2 numbers in this paper are quoted from those runs.

5.4 Cross-implementation check

The browser version of the model is an independent implementation in a different language with a different random-number generator. On the Phase-2 default-fleet cells (four mechanisms, six metrics, 30 seeds each), every one of the 24 metric means lies within the combined 95% confidence intervals of the two implementations. Agreement of this kind does not prove either implementation right, but it rules out the class of bugs that a single implementation hides from itself.

Compute. All runs were made on a 12-core desktop under Windows 11. One simulated year takes 0.1–3 s depending on demand; the fourteen suites, about 8,000 runs plus the sequential-replication runs, took about 40 minutes on 10 worker processes. Discarded preliminary runs during development amounted to a few thousand.

Table 4. Experiment suites. Fleets are listed as owner:class; LTVs are supervised, haulers supervised or autonomous, the small rover teleoperated unless stated.

Suite	Question	Factors and levels	Mechanisms	Cells × seeds
E1 Phases	Main comparison	Phase 1 (3 vehicles), 2 (6), 3 (9) default fleets; plus an unconstrained 16-vehicle autonomous fleet as a bound	all four + bound	15 × 30
E2 Fleet size	Minimum fleet for service level	Phase-2 demand; fleets of 2, 3, 4, 5, 6, 8 vehicles	all four	24 × 30
E3 Thickness	Pooling gain vs fragmentation	6 vehicles split among 1, 2, 3, 4, 6 owners; other-duty share 0, 0.3, 0.6	in-house, marketplace	30 × 30
E4 Heavy cargo	The capability gap	Phases 2 and 3; fleets of 4 LTVs, 3 LTVs + 1 hauler, 2 LTVs + 2 haulers	greedy, marketplace, marketplace without team lift	18 × 30
E5 Autonomy and latency	Throughput bounds	All-teleop, all-supervised, all-autonomous 5-vehicle fleets; RTT 2.6, 5, 8, 16, 603, 1,323 s; 1, 2, 4 operators	marketplace	54 × 20
E6 Standards	Interface compatibility	Share of cargo on the common standard 0.5, 0.75, 0.9, 1.0; 3 of 5 vehicles accept only the common standard	in-house, greedy, marketplace	12 × 30
E7 Environment	Robustness	Dark share 0.1–0.5; MTBF ×0.25–×2; bulk 0–3,000 t/yr	greedy, marketplace	24 × 30
E8 Ablations	Which components matter	Each of seven components removed; batching window 0, 0.5, 1, 2, 6 h; Phases 2 and 3 with tight fleets	marketplace variants	24 × 30
E9 Tuning	Do tuned weights transfer?	40 random weight configurations on Phase 2 (8 seeds); best vs hand-set on Phases 1–3 (30 seeds)	marketplace	41 × 8 + 6 × 30
E10 Economics	Cost drivers	Demand ×0.5/×1.5; operator cost; other duty; dark share; MTBF; plus analytic vehicle cost and life	marketplace	11 × 30
E11 Anticipation	Does staging before landings help?	Phases 1–3 tight fleets; Phase 2 with 2 and 3 vehicles	marketplace with and without anticipation	10 × 30
E12 Capacity ladder	What capacity moves the manifest?	300 manifest-years per phase; capacities 300–30,000 kg	manifest analysis, no dispatch	3 × 300
E13 Fragmentation law	Does in-house service depend only on vehicles per owner?	Fleets of 4, 6, 8 LTVs; 1–8 owners; duty 0, 0.3, 0.6	in-house, marketplace	90 × 30
E14 Precision	Replicate until settled	Phase-2 default fleet; 99% CI within 1% (5% for p90)	all four	sequential, up to 4,000 each

6. Results

6.1 Main comparison across phases

Pooling of any kind beats bilateral contracting, in every phase. Figure 4 and Table 5 give the main comparison with each phase's default fleet (3, 6 and 9 vehicles). In-house fleets deliver urgent cargo with a 90th-percentile delay of 42 h in Phase 1, 49 h in Phase 2 and 23 h in Phase 3; the marketplace brings those to 27, 15 and 8.5 h, paired differences of -15.3 ± 5.9 , -33.7 ± 12.0 and -14.1 ± 5.3 h (95% CI over 30 seeds). The share of urgent items delivered within a day rises by 16, 13 and 8 absolute points. The in-house rule also leaves 4–12 tonnes of landed cargo undelivered at year end in Phases 2 and 3, because items heavier than 1.5 t belong to owners whose contracted carrier has no hauler while another company's hauler sits idle.

Among pooled rules the ordering is consistent but needs many replications to resolve. With 30 seeds the marketplace's advantage over pooled greedy on the 90th-percentile urgent delay is -2.9 ± 6.5 h in Phase 1, -4.9 ± 4.6 h in Phase 2 and -1.1 ± 1.1 h in Phase 3; against batched assignment it is 0.0 ± 0.6 , -3.3 ± 3.9 and -0.8 ± 1.1 h. Only the Phase-2 difference against greedy is distinguishable from zero at that sample size. Section 6.14 replicates the Phase-2 cells until their 99% confidence intervals settle (4,000–53,000 replications per cell): the settled mean urgent delays are 14.62 ± 0.15 h in-house, 7.44 ± 0.07 h pooled greedy, 7.54 ± 0.25 h batched assignment and 6.31 ± 0.22 h marketplace (99% CIs), so the marketplace is 15% faster than greedy pooling on the mean and 14% on the 90th percentile (15.1 against 17.6 h), with a same-day service level of 0.938 against 0.920. The unconstrained 16-vehicle autonomous fleet delivers urgent items in 2.0–2.4 h on average, which is essentially travel and handling time; default fleets therefore spend most of the urgent delay waiting for a vehicle to be free, lit and off other duty, not driving.

Where the marketplace differs most is in what it can move and when it drives. With LTV-class rovers and one hauler, 10–17% of landed cargo mass in every phase is heavier than any single vehicle. The marketplace's team lifts cut that to 0% in Phases 1 and 3 and 2.8% in Phase 2 (the residual is 13–15 t elements that even a hauler plus two LTVs cannot carry together). Vehicles without night capability are caught by the lunar night 0.2, 1.3 and 4.1 times a year under greedy dispatch and 0, 0.13 and 0.17 times under the marketplace, each event costing a job about 106 h. Bundling cuts trips by 4–7%. The price is a deliberate one: deferrable bulk lots wait 17–35 h on average under consolidation against 3–12 h under greedy dispatch. Operator hours fall slightly in Phases 1 and 2 and rise by 263 ± 29 h (34%) in Phase 3, where the generalised-cost bid lets cheap, slow, remotely operated vehicles win bulk lots; Section 6.9 shows that tuned weights reduce this increase.

Pooling also spreads the work. The Gini coefficient of revenue across vehicle owners is 0.49–0.72 under in-house contracting and 0.20–0.46 under the pooled rules. Empty kilometres are about half of all kilometres under every rule, far above the 16–21% deadhead of Earth trucking, because almost all cargo flows one way, from pads to base; return items are too few to change this.

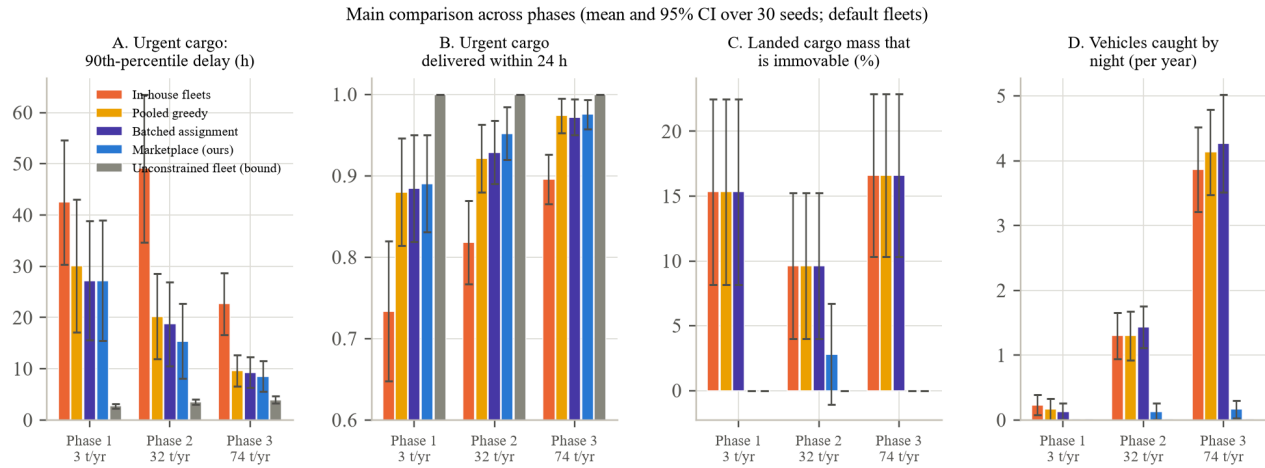


Figure 4. Main comparison with each phase's default fleet. Bars are means over 30 seeds; whiskers are 95% confidence intervals. **A:** 90th-percentile delay of urgent items from release to delivery. **B:** share of urgent items delivered within 24 h. **C:** share of landed cargo mass that no vehicle or team in the fleet could ever move. **D:** number of times per year a vehicle without night capability is caught out by the lunar night. The grey bars are a 16-vehicle fully autonomous fleet with no other duties or faults, included as a bound.

Table 5. Main results with default fleets: mean $\pm$ 95% CI over 30 seeds. Urgent = priority-1 items. Immovable = share of landed cargo mass no vehicle or team could ever carry. Bulk delay is the mean delay of deferrable regolith lots, which the marketplace holds on purpose. Cost is amortised fleet plus operator time per kilogram of all mass moved.

Phase	Mechanism	Urgent p90 delay (h)	Urgent mean delay (h)	Urgent within 24 h	Immovable (%)	Night strandings / yr	Trips / yr	Bulk delay (h)	Operator h / yr	Revenue Gini	Cost \$/kg moved
Phase 1	In-house fleets	42.5 $\pm$ 12.2	19.9 $\pm$ 6.5	0.73 $\pm$ 0.09	15.3	0.23 $\pm$ 0.15	78	19.1	61	0.49	3,052
Phase 1	Pooled greedy	30.1 $\pm$ 13.0	12.6 $\pm$ 5.4	0.88 $\pm$ 0.07	15.3	0.17 $\pm$ 0.17	79	11.8	64	0.35	3,015
Phase 1	Batched assignment	27.2 $\pm$ 11.6	12.5 $\pm$ 5.2	0.88 $\pm$ 0.07	15.3	0.13 $\pm$ 0.12	79	11.8	59	0.34	3,015
Phase 1	Marketplace (ours)	27.2 $\pm$ 11.7	11.5 $\pm$ 4.7	0.89 $\pm$ 0.06	0.0	0.00 $\pm$ 0.00	76	35.4	57	0.35	2,935
Phase 1	Unconstrained 16-vehicle bound	2.7 $\pm$ 0.5	2.0 $\pm$ 0.2	1.00 $\pm$ 0.00	0.0	0.00 $\pm$ 0.00	79	26.5	7	0.00	22,435
Phase 2	In-house fleets	49.1 $\pm$ 14.4	16.2 $\pm$ 4.5	0.82 $\pm$ 0.05	9.6	1.30 $\pm$ 0.35	548	12.3	361	0.64	489
Phase 2	Pooled greedy	20.2 $\pm$ 8.4	7.6 $\pm$ 2.3	0.92 $\pm$ 0.04	9.6	1.30 $\pm$ 0.38	554	4.3	479	0.46	478
Phase 2	Batched assignment	18.7 $\pm$ 8.2	7.5 $\pm$ 2.1	0.93 $\pm$ 0.04	9.6	1.43 $\pm$ 0.32	554	4.6	431	0.44	478
Phase 2	Marketplace (ours)	15.4 $\pm$ 7.3	5.9 $\pm$ 1.7	0.95 $\pm$ 0.03	2.8	0.13 $\pm$ 0.12	529	23.4	459	0.45	475
Phase 2	Unconstrained 16-vehicle bound	3.5 $\pm$ 0.6	2.2 $\pm$ 0.2	1.00 $\pm$ 0.00	0.0	0.00 $\pm$ 0.00	542	22.3	54	0.00	1,465
Phase 3	In-house fleets	22.7 $\pm$ 6.0	9.4 $\pm$ 1.5	0.90 $\pm$ 0.03	16.6	3.87 $\pm$ 0.65	1692	7.3	894	0.72	267
Phase 3	Pooled greedy	9.6 $\pm$ 3.1	5.0 $\pm$ 1.0	0.97 $\pm$ 0.02	16.6	4.13 $\pm$ 0.66	1694	3.2	766	0.22	266
Phase 3	Batched assignment	9.3 $\pm$ 3.0	5.0 $\pm$ 1.0	0.97 $\pm$ 0.02	16.6	4.27 $\pm$ 0.76	1694	3.5	765	0.23	266
Phase 3	Marketplace (ours)	8.5 $\pm$ 3.0	4.2 $\pm$ 0.8	0.98 $\pm$ 0.02	0.0	0.17 $\pm$ 0.14	1569	16.9	1029	0.20	264
Phase 3	Unconstrained 16-vehicle bound	4.0 $\pm$ 0.7	2.4 $\pm$ 0.2	1.00 $\pm$ 0.00	0.0	0.00 $\pm$ 0.00	1602	16.1	181	0.00	493

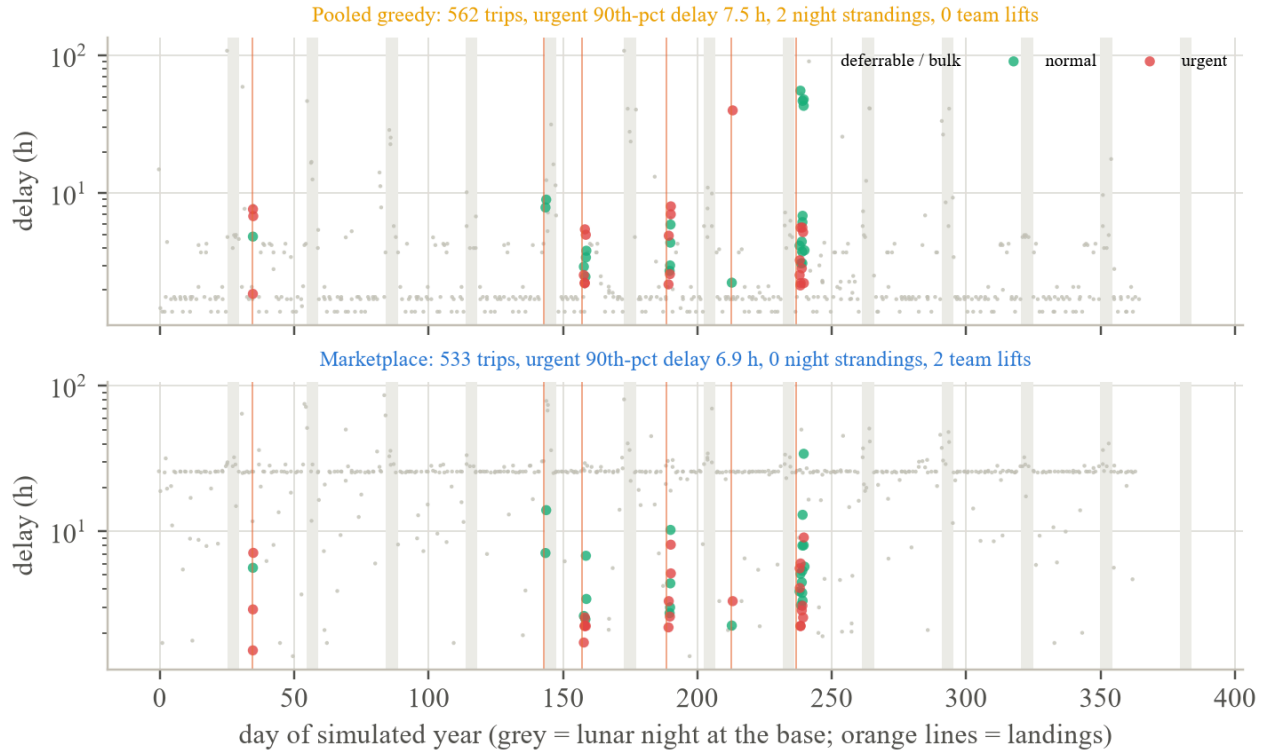


Figure 5. One simulated Phase-2 year (seed 3, four-vehicle fleet) under pooled greedy dispatch (top) and the marketplace (bottom). Each dot is a delivered item at its release day and its delay. Grey bands are lunar nights at the base, orange lines are landings. The marketplace's deferrable lots cluster at the 24 h consolidation limit; its urgent items (red) are served at the same speed as under greedy dispatch, and no vehicle is caught by the night. The same replay, animated on the base map, is on the Replay tab of the browser version.

6.2 How small a fleet can serve Phase 2?

Because the fleet is the cost, the practical question is not which rule delivers fastest with a given fleet, but which rule meets a service target with the fewest vehicles. Figure 6 sweeps fleets of 2 to 8 vehicles at Phase-2 demand. Every pooled rule reaches a 90% same-day service level for urgent cargo with four vehicles (marketplace 0.940 ± 0.030 , batched assignment 0.927 ± 0.036 , pooled greedy 0.921 ± 0.036); in-house fleets peak at 0.87 with eight. In annual fleet cost at our unit prices, four vehicles are \$162 million a year and eight are \$357 million, so pooling is worth at least \$195 million a year at this demand level, and in-house contracting does not buy the service level at any fleet size we tried. Below four vehicles no rule reaches the target: with two LTVs the 90th-percentile urgent delay is 47–48 h for the pooled rules and 67 h in-house, because a single landing can release more items than two rovers can clear before the next night. Among the pooled rules the marketplace has the lowest 90th-percentile delay at four, six and eight vehicles (16.9, 15.4 and 9.3 h against 19.4, 20.2 and 11.0 h for greedy), but the differences at 30 seeds are within confidence intervals; the settled Phase-2 estimates of Section 6.14 confirm the ordering.

System cost per kilogram of all mass moved (including regolith) rises almost linearly with fleet size, from about \$175 with 2 vehicles to about \$675 with 8, because demand is fixed and the fleet is 99% of the cost. Per kilogram of *landed* cargo the picture depends on what gets moved: with two LTVs the marketplace delivers heavy items by team lift that greedy dispatch leaves stranded, so its landed-kg cost is \$21,000 against \$24,000

for greedy and \$26,000 in-house; with four vehicles the pooled rules are at \$34,000 per landed kilogram and in-house at \$71,000, because in-house leaves 11 t undelivered.

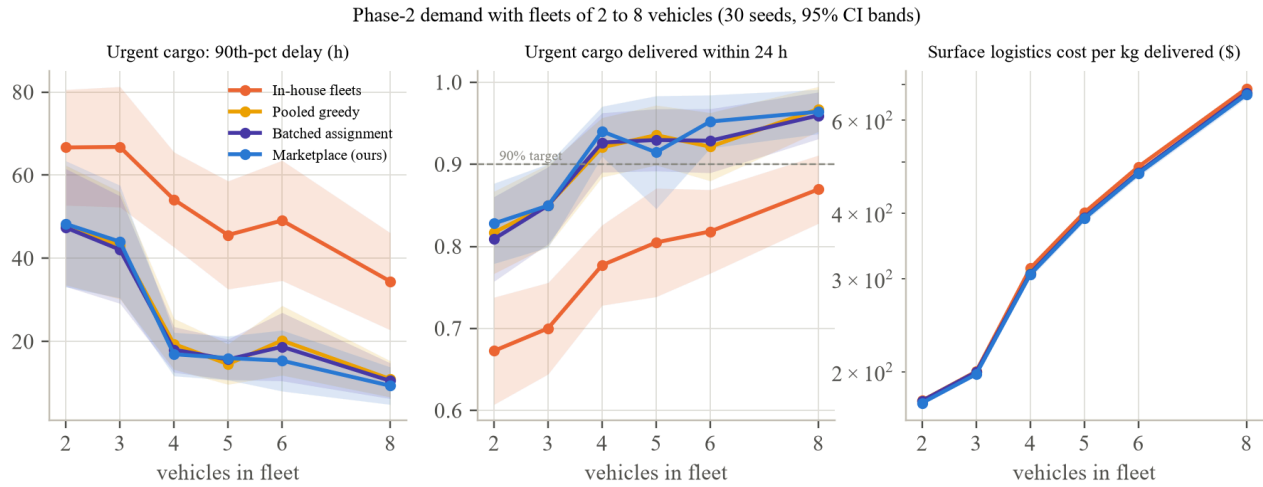


Figure 6. Fleet-size sweep at Phase-2 demand (30 seeds; bands are 95% CIs). Fleets grow from two LTVs to eight mixed vehicles (Table 4). **Left:** urgent 90th-percentile delay. **Middle:** share of urgent items delivered within 24 h with the 90% target marked. **Right:** system cost per kilogram of all mass moved; the curves lie on top of each other because the fleet, not the dispatch rule, sets the cost.

6.3 Market thickness: when does pooling pay?

Figure 7 holds the fleet fixed at six vehicles and splits it among one to six owners, with each owner's cargo contracted to its own vehicles under the in-house rule. With a single owner the in-house rule and the marketplace are statistically indistinguishable: the paired gain in urgent 90th-percentile delay is 0.7 ± 3.0 h with no other duties, 3.9 ± 4.1 h at 30% other duties and 3.3 ± 6.0 h at 60%. The gain grows monotonically with fragmentation. With six owners it is 23 ± 9 h, 44 ± 8 h and 52 ± 11 h at the three duty levels; the in-house same-day service level falls from 0.96 to 0.88, 0.72 and 0.51 while the marketplace's stays at 0.96, 0.955 and 0.91 regardless of how many owners there are. Other duties amplify the effect because a vehicle committed to its owner's science traverse cannot serve its owner's cargo, while under pooling a neighbour's idle vehicle can.

This is the thin-market result we expected from Earth freight, but in a sharper form. The lunar market will have very few participants, so whether pooling matters depends almost entirely on how many of them there are. If NASA's two LTV providers and one or two commercial payload companies each run their own vehicles for their own cargo, the simulations say that a shared dispatch layer roughly halves urgent-cargo delay and keeps every owner's heavy items moving; if a single integrator owns the whole fleet, the dispatch layer adds little that a good in-house scheduler would not. Section 6.13 turns this into a quantitative law.

Market thickness: pooling gain grows with fragmentation and with vehicles' other duties

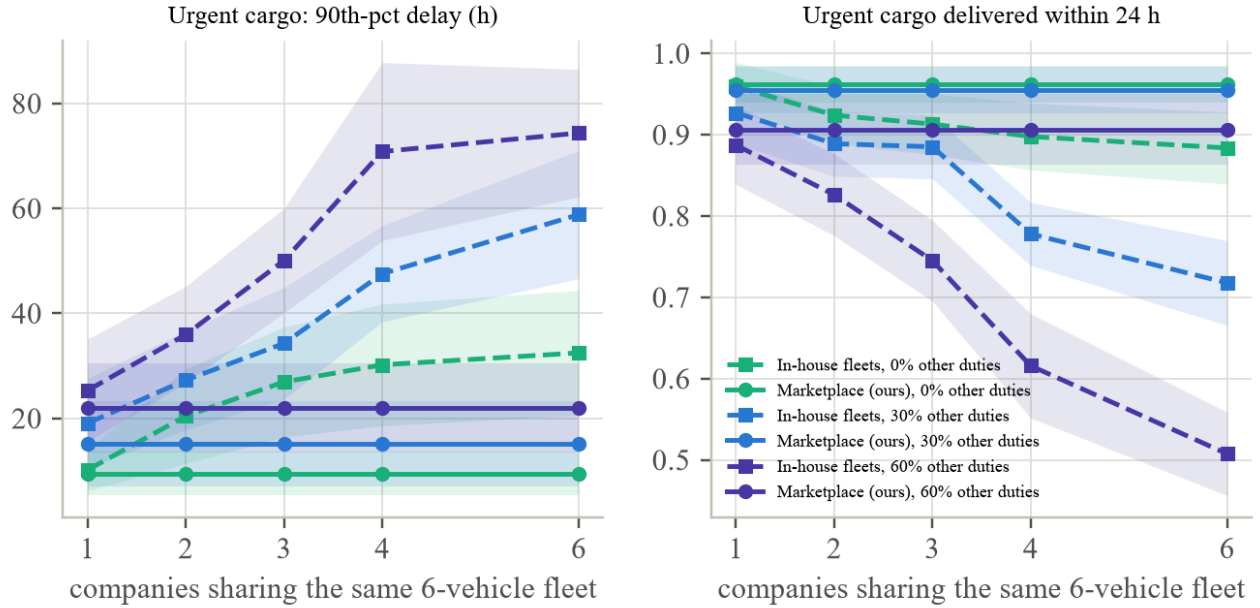


Figure 7. Market thickness. Six vehicles (five LTVs and one hauler) are split among 1, 2, 3, 4 or 6 owners; cargo is drawn from the same owners. Dashed lines with squares are in-house contracting, solid lines with circles the marketplace; colours give the share of time vehicles spend on their owners' other duties. Bands are 95% CIs over 30 seeds. The marketplace lines are flat because pooling makes ownership irrelevant to dispatch.

6.4 The heavy-cargo gap

NASA's statement that no planned vehicle can relocate a large element translates, in the simulations, into a large share of immovable mass. With four LTV-class rovers and no hauler, $54 \pm 8\%$ of landed cargo mass in Phase 2 and $69 \pm 5\%$ in Phase 3 is heavier than any vehicle (Figure 8). Team lifts by two or three LTVs, which can carry up to 3 and 4.5 t together, recover about half of that: the immovable share falls to $29 \pm 7\%$ and $42 \pm 7\%$. Replacing one LTV with a 10-tonne hauler leaves 10% and 17% immovable under greedy dispatch (the 12–15 t elements), and 2.8% and 7.5% under the marketplace, where a hauler and two LTVs lift together. Two haulers remove the gap entirely in both phases. Items that *can* be moved are not slowed by the team lifts: the urgent 90th-percentile delay with one hauler is 14.8 h under the marketplace against 15.9 h under greedy dispatch in Phase 2.

We read this as follows. Coordination software can stretch a fleet of 1.5-tonne rovers to the 2–3 tonne items that MK1-class landers deliver, which is most of the Phase-2 manifest by count. It cannot reach the 6–15 tonne elements that define Phase 3. One heavy vehicle, dispatched as part of a shared pool so that it also lends its capacity to others' items, closes most of the remaining gap; a second closes all of it. Section 6.12 generalises this into a capacity ladder that does not depend on the fleet we happened to simulate.

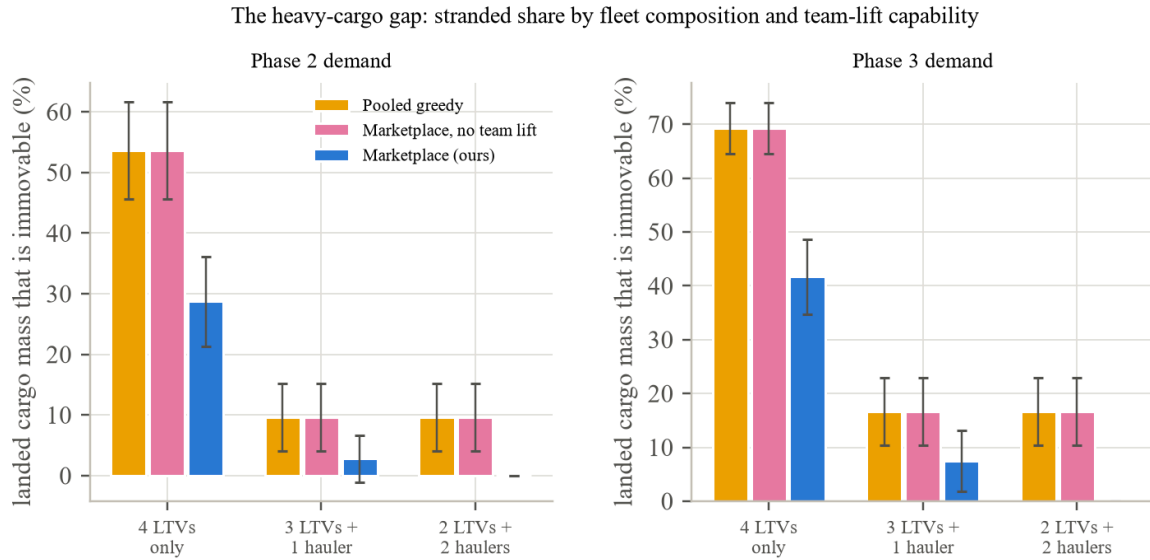


Figure 8. Share of landed cargo mass that no vehicle or team could move, by fleet composition and phase (30 seeds; 95% CIs). Team lift (blue) recovers about half of the mass stranded by an LTV-only fleet and most of the mass stranded with a single hauler.

6.5 Autonomy and latency

Figure 9 compares a five-vehicle Phase-2 fleet operated entirely by Earth teleoperation, supervised autonomy or full autonomy, across round-trip latencies from 2.6 s (direct light time) to 1,323 s (Mars at its farthest). With supervised or full autonomy the latency does not matter in our model: urgent 90th-percentile delay is 12.1 and 11.9 h at every latency, and operator hours are 342 and 63 per year. Teleoperation is workable at lunar latency but costly: with two operators the urgent 90th-percentile delay is 29 h at 2.6 s, 38 h at the 5 s default and 48 h at 8 s, rising to 104 h at a relayed 16 s; operator consoles are staffed 2,100–5,900 hours a year, that is one to three full-time positions for five rovers. A single operator for five teleoperated rovers is a bottleneck in its own right: jobs wait for an operator 280–580 times a year and the 90th-percentile delay roughly doubles relative to two operators. At Mars-like latencies teleoperated speed falls to a few tens of metres per hour and the fleet delivers only 2–16% of the year's mass.

Two conclusions follow for a marketplace that dispatches other people's vehicles. First, the vehicles' autonomy level is a market-design variable: a pool of teleoperated rovers needs the marketplace to schedule operators as well as vehicles, and even then runs 2–3 times slower on urgent cargo. Second, nothing that works by teleoperation on the Moon transfers to Mars; the Mars column in Figure 9 is a different problem.

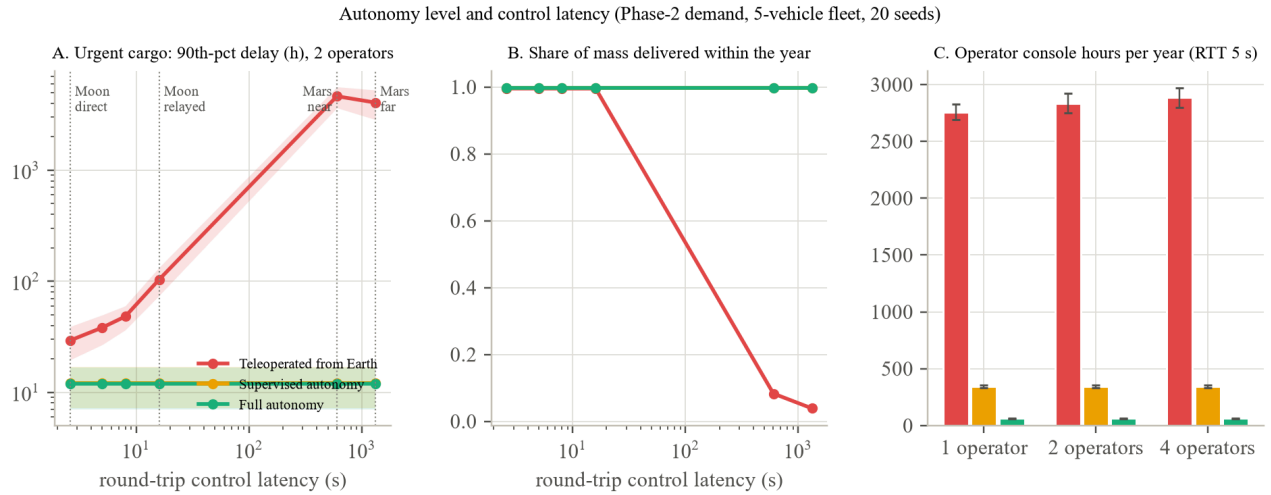


Figure 9. Autonomy level and control latency at Phase-2 demand with a five-vehicle fleet (20 seeds; bands are 95% CIs). **A:** urgent 90th-percentile delay with two Earth operators, on log axes; the dotted lines mark direct and relayed lunar latency and the near and far Mars cases. **B:** share of the year's mass delivered. **C:** operator console hours per year at the 5 s default latency for one, two and four operators.

6.6 Interface standards

In this suite three of five vehicles accept only the common cargo standard and two accept both, while the share of landed cargo built to the common standard varies from 50% to 100% (Figure 10). Nothing becomes immovable, because every item has at least two compatible vehicles, but delays grow as compatibility shrinks, and they grow most for the rules that cannot route around it: at 50% common-standard cargo the in-house urgent 90th-percentile delay is 54 h against 46 h at full compatibility, pooled greedy's is 22 h against 15 h, and the marketplace's is 16.9 h against 16.0 h. The service level for urgent items moves from 0.75 to 0.81 in-house, from 0.91 to 0.94 under greedy pooling and stays at 0.92–0.93 under the marketplace. The marketplace is the least sensitive because its auction prices the two flexible vehicles into the jobs only they can do, whereas greedy dispatch spends them on whatever is nearest.

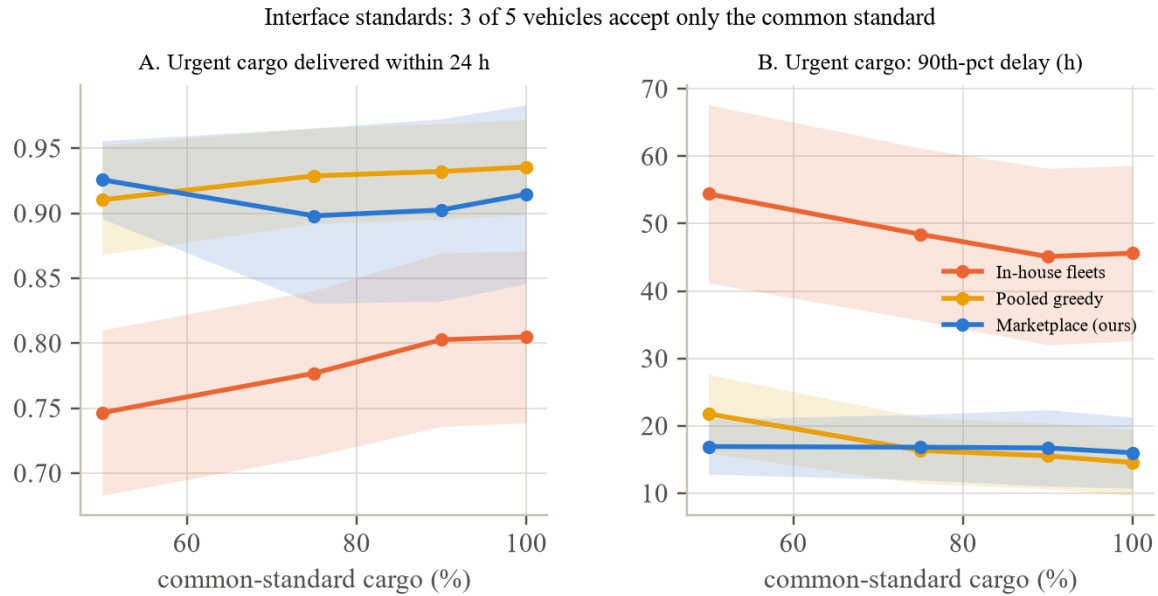


Figure 10. Interface standards at Phase-2 demand (30 seeds; 95% CIs). Three of five vehicles accept only the common standard. **A:** urgent same-day service level. **B:** urgent 90th-percentile delay.

6.7 Night, reliability and bulk flows

Figure 11 sweeps three environment factors with a five-vehicle Phase-2 fleet. **Night.** Delay grows steeply with the dark share of the cycle: at 30% dark the urgent 90th-percentile delay is 47 h under greedy dispatch and 30 h under the marketplace (paired difference -18 ± 10 h); at 50% dark, 63 against 58 h (-5 ± 12 h, not distinguishable from zero because at that point half the month is lost to everyone). The difference is night awareness: greedy dispatch sends vehicles out that cannot return before dark 1.1–1.3 times a year regardless of the dark share, and each such event removes the vehicle and its cargo for the whole night. **Reliability.** Quartering the MTBF (to about 125 h for an LTV, 12 faults a year across the fleet) raises the greedy 90th-percentile delay from 15 to 21 h and the marketplace's from 16 to 19 h; the paired difference, -1.5 ± 3.9 h, is not distinguishable from zero. Faults matter less than nights because recovery takes 12–36 h rather than 106. **Bulk flows.** Raising regolith and feedstock flows from zero to 3,000 t a year raises utilisation from 0.5% to 16–17% and leaves urgent delay roughly unchanged until the top level, where both rules degrade to about 34 h; the marketplace makes 5–14% fewer trips, more as the flows grow, and the greedy rule's night strandings rise to 7.6 a year. Even at 3,000 t a year the fleet is idle five sixths of the time, which is a reminder that these vehicles will be bought for capability, not for throughput.

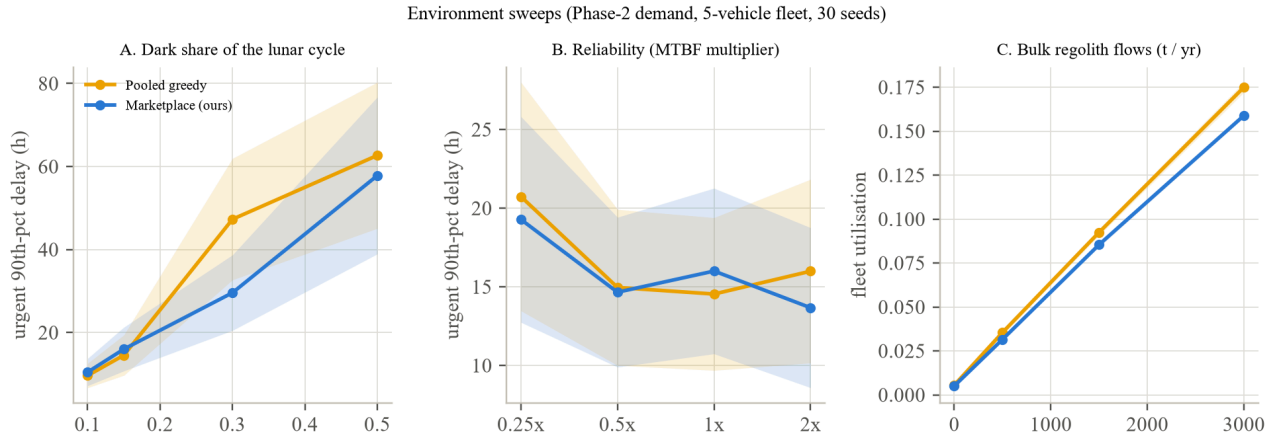


Figure 11. Environment sweeps at Phase-2 demand with a five-vehicle fleet (30 seeds; bands are 95% CIs). **A:** dark share of the lunar cycle at the base. **B:** mean time between faults as a multiple of the default. **C:** bulk regolith and feedstock flows, shown as fleet utilisation.

6.8 Ablations and the batching window

Figure 12 and Table 6 remove one marketplace component at a time with tight fleets (four vehicles in Phase 2, six in Phase 3). Two components carry the urgent-delay result in Phase 3, where cargo is dense enough for them to act: without bundling the 90th-percentile delay rises by 4.6 ± 4.4 h with 131 more trips a year, and without consolidation there are 60 more trips; both removals also *reduce* operator hours by 340–370 a year, because consolidated loads tend to go to the slower, remotely operated vehicles that win them on price. Night awareness carries the night result: removing it brings back 1.0 ± 0.3 and 5.8 ± 1.2 night strandings a year and adds 36 ± 12 and 275 ± 60 operator hours (hibernating vehicles keep their operators booked). Team lift carries the stranded-mass result and costs nothing measurable on urgent delay; in Phase 3 its removal improves urgent delay by 1.6 ± 2.6 h because vehicles are no longer tied up in pairs. Three components did nothing measurable in these scenarios: anticipatory staging ($+0.3 \pm 0.5$ and -1.4 ± 1.5 h when removed; Section 6.11 explains why), the priority credit ($+0.2 \pm 0.3$ and $+1.4 \pm 3.8$ h) and backhaul (0.0 and $+0.2 \pm 0.8$ h). The value-of-time term in the bid already orders urgent items first, and return items are rare enough that a vehicle seldom finds one at its last stop. We keep all three in the mechanism for completeness but make no claim for them.

The batching window behaves as the thin-market theory predicts [11]. Dispatching on every event is best in both phases; windows of 0.5–2 h add 1–5 h to the urgent 90th-percentile delay, and a 6 h window adds 9 h in Phase 2 and 21 h in Phase 3 while bundling only 10–15 more trips. There is nothing to gain from waiting when the next vehicle to free up is hours away and the next item to arrive is days away.

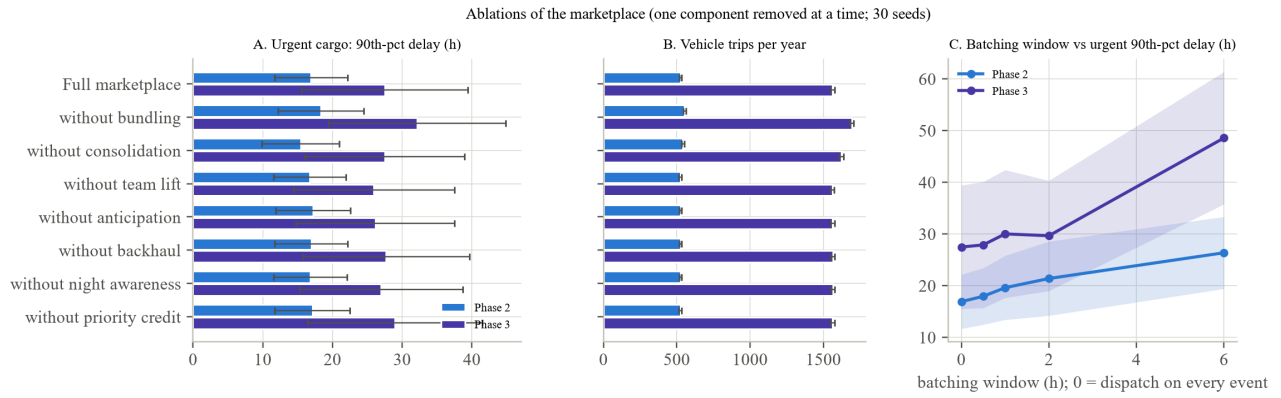


Figure 12. Ablations with tight fleets (four vehicles in Phase 2, six in Phase 3; 30 seeds; 95% CIs). **A:** urgent 90th-percentile delay with each component removed. **B:** vehicle trips per year. **C:** urgent delay as a function of the batching window; zero means dispatch on every event.

Table 6. Ablations with tight fleets (Phase 2: four vehicles; Phase 3: six vehicles). Each cell is the mean over 30 seeds; the bracketed value is the paired difference from the full marketplace with its 95% CI. Immovable = share of landed cargo mass no vehicle or team could carry.

Phase	Variant	Urgent p90 delay (h)	Trips / yr	Night strandings / yr	Immovable (%)	Bulk delay (h)	Operator h / yr
Phase 2	Full marketplace	16.9	524	0.10	2.8	24.7	402
Phase 2	without bundling	18.3 [+1.4 ± 2.4]	554 [+30 ± 2]	0.10 [+0.00 ± 0.16]	2.8	4.3 [-20.4 ± 0.1]	402 [-1 ± 7]
Phase 2	without consolidation	15.4 [-1.5 ± 2.4]	542 [+18 ± 2]	0.07 [-0.03 ± 0.15]	2.8	4.3 [-20.4 ± 0.1]	396 [-6 ± 7]
Phase 2	without team lift	16.7 [-0.2 ± 0.2]	524 [-1 ± 0]	0.07 [-0.03 ± 0.07]	9.6	24.7 [-0.0 ± 0.0]	398 [-4 ± 3]
Phase 2	without anticipation	17.2 [+0.3 ± 0.5]	524 [+0 ± 0]	0.07 [-0.03 ± 0.11]	2.8	24.7 [+0.0 ± 0.0]	404 [+2 ± 4]
Phase 2	without backhaul	16.9 [+0.0 ± 0.0]	524 [+0 ± 0]	0.10 [+0.00 ± 0.00]	2.8	24.7 [-0.0 ± 0.0]	402 [-0 ± 0]
Phase 2	without night awareness	16.9 [-0.1 ± 0.9]	525 [+0 ± 0]	1.13 [+1.03 ± 0.30]	2.8	24.9 [+0.2 ± 0.1]	438 [+36 ± 12]
Phase 2	without priority credit	17.1 [+0.2 ± 0.3]	524 [-0 ± 0]	0.10 [+0.00 ± 0.00]	2.8	24.7 [-0.0 ± 0.0]	403 [+0 ± 1]
Phase 3	Full marketplace	27.5	1563	0.30	7.5	17.8	1839
Phase 3	without bundling	32.1 [+4.6 ± 4.4]	1694 [+131 ± 4]	0.33 [+0.03 ± 0.22]	7.5	5.9 [-12.0 ± 0.2]	1500 [-339 ± 20]
Phase 3	without consolidation	27.5 [-0.0 ± 5.7]	1623 [+60 ± 6]	0.37 [+0.07 ± 0.26]	7.5	6.0 [-11.9 ± 0.1]	1467 [-373 ± 18]
Phase 3	without team lift	25.9 [-1.6 ± 2.6]	1560 [-3 ± 1]	0.20 [-0.10 ± 0.14]	16.6	17.8 [-0.1 ± 0.1]	1802 [-37 ± 13]
Phase 3	without anticipation	26.2 [-1.3 ± 1.5]	1563 [-0 ± 2]	0.27 [-0.03 ± 0.18]	7.5	17.9 [+0.0 ± 0.1]	1849 [+9 ± 14]
Phase 3	without backhaul	27.7 [+0.2 ± 0.8]	1564 [+1 ± 1]	0.27 [-0.03 ± 0.07]	7.5	17.8 [-0.0 ± 0.0]	1839 [-1 ± 4]
Phase 3	without night awareness	27.0 [-0.5 ± 6.5]	1564 [+1 ± 3]	6.13 [+5.83 ± 1.24]	7.5	18.1 [+0.3 ± 0.1]	2114 [+275 ± 60]
Phase 3	without priority credit	28.9 [+1.4 ± 3.8]	1562 [-1 ± 3]	0.37 [+0.07 ± 0.16]	7.5	17.8 [-0.0 ± 0.1]	1843 [+3 ± 12]

6.9 Do tuned weights transfer?

We searched 40 random weight configurations on the Phase-2 tight-fleet scenario with eight seeds and tested the best against the hand-set weights on all three phases with 30 fresh seeds (Figure 13). Across the 40 configurations the urgent 90th-percentile delay on the search set ranged from 8.0 to 12.6 h with a mean of 9.5 h; the hand-set configuration scored 11.5 h, in the bottom half. The best configuration uses a much larger urgency credit for both priority classes (29 h and 6.6 h against 6 h and 2 h), a stronger age credit, and lets bulk lots wait longer (up to 63 h). On the test seeds it improves the urgent 90th-percentile delay by 3.1 ± 1.6 h in Phase 2 and 7.4 ± 7.3 h in Phase 3, leaves Phase 1 unchanged (-0.4 ± 0.5 h), and cuts operator hours by 15 ± 6 and 96 ± 20 a

year; the cost is longer bulk-lot waits (+8 to +31 h). The same-day service level moves by at most one point. We take two things from this. The mechanism's urgent-cargo performance is set mainly by its structure (pooling, team lift, night awareness, event-driven dispatch) and only secondarily by its weights, which buy a further 10–25% on the tail; and the weights that matter trade bulk-lot patience against urgent-cargo speed and operator time, which is a policy choice an operator can make per base rather than a parameter to fit.

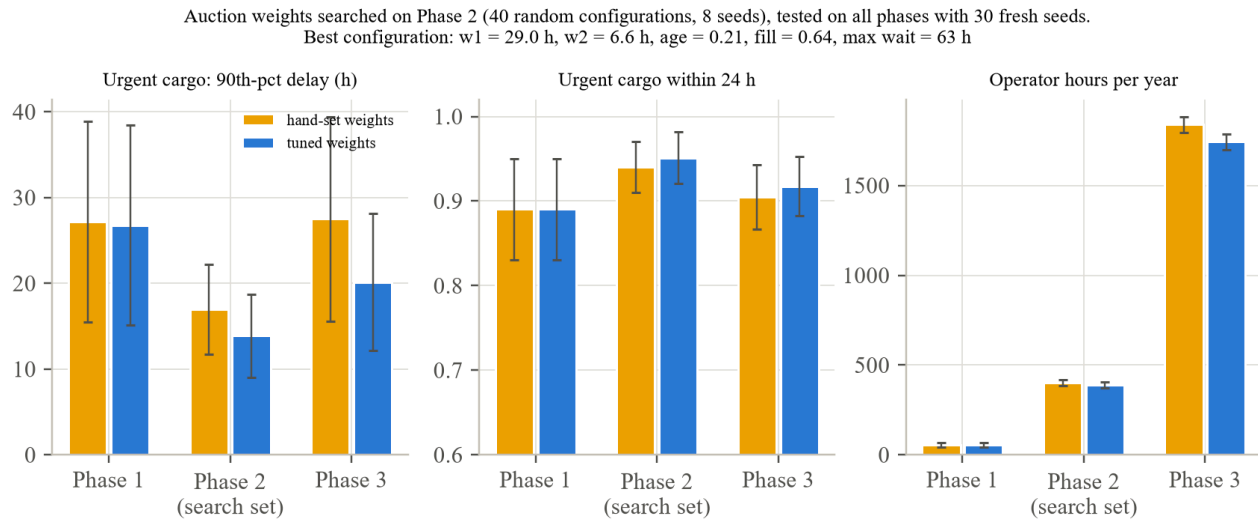


Figure 13. Hand-set versus tuned auction weights (30 fresh seeds per phase; 95% CIs). Weights were searched on Phase 2 only. The tuned weights shorten the urgent tail in Phases 2 and 3 and reduce operator hours, at the cost of longer waits for deferrable bulk lots.

6.10 What a surface move costs

Figure 14 puts the cost side together for the Phase-2 four-vehicle marketplace cell. The annual bill is \$162 million of amortised fleet and \$0.4 million of operator time. Divided by the 532 t moved in the year (32 t of landed cargo and 500 t of regolith and feedstock) that is \$306 per kilogram moved; divided by landed cargo alone it is \$5,100 per kilogram, and \$5,000 in Phase 3 with the nine-vehicle default fleet. The tornado in panel A shows that only two assumptions move this number: vehicle unit cost ($\pm 50\%$ gives \$153–458 per kg) and vehicle life (15 years or 6.7 years gives \$204–458). Halving or multiplying demand by 1.5 changes it by 3%; operator cost by 1%; other duties, dark share and reliability by nothing, because they change delay, not cost. Panel B puts the surface move next to the cost of getting the kilogram to the surface: about \$1.08 million per kg on a CLPS small lander (Blue Ghost Mission 1), about \$78,000 on a fully loaded Blue Moon MK1 and SpaceX's published \$100,000 for Starship. Moving a kilogram across the base costs 0.5–6.5% of landing it.

Three things follow for anyone pricing the service. The marginal cost of a job is small and the fixed cost is enormous, so prices will be set by capacity commitments rather than by job costs, as in contract trucking rather than the spot market. Dispatch quality shows up as fleet size, not operating cost: the difference between a four-vehicle pooled fleet and an eight-vehicle in-house fleet that still misses the target is the economic content of the mechanism comparison. And the service is cheap relative to what it protects: a day's delay on a \$1 million-per-kilogram crate is worth far more than the few thousand dollars it costs to move it, which is why urgent-cargo delay rather than cost is the right primary metric for this market.

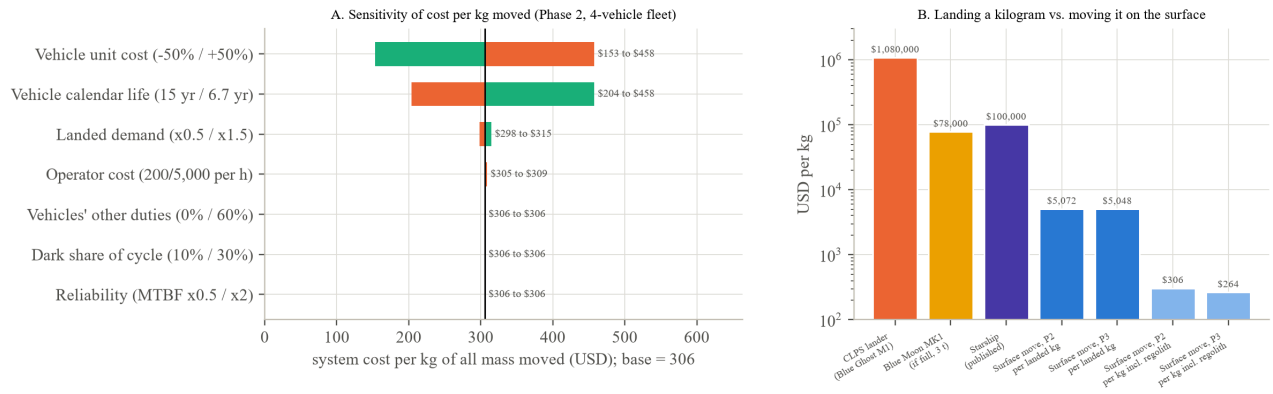


Figure 14. Economics. **A:** tornado of system cost per kilogram moved for the Phase-2 four-vehicle marketplace cell; vehicle unit cost and life are analytic swings on the cost model, the others are simulated (30 seeds). **B:** cost of landing a kilogram on three vehicles [46, 47, 4] against the simulated cost of moving it on the surface, per landed kilogram and per kilogram of all mass moved including regolith.

6.11 Anticipatory staging: a sensible idea that does not help here

Landings are published months ahead, so a marketplace can send an idle vehicle to the pad before touchdown. We expected a gain and found none (Figure 17). Across five fleet-and-phase cells the paired change in urgent 90th-percentile delay from switching anticipation on is between -0.7 ± 0.7 h and $+1.3 \pm 1.5$ h, and in mean delay between -0.4 ± 0.5 and $+0.2 \pm 0.2$ h; the same-day service level changes by at most one point. The vehicles make 3–10 staging moves a year and drive 8–31 extra empty kilometres. Three facts explain the null result. The base is compact: a vehicle at the depot reaches either pad in 20–30 minutes, which is small against offload windows of 6–72 h. Vehicles are idle most of the time, so one is usually free when the first item is ready. And the first items off a lander are rarely the urgent ones that dominate the tail. Anticipation would matter at a base whose depot is many kilometres from its pads, or at utilisations far above anything Phase 2 produces; at this base it is a cost without a benefit, and we report it as such.

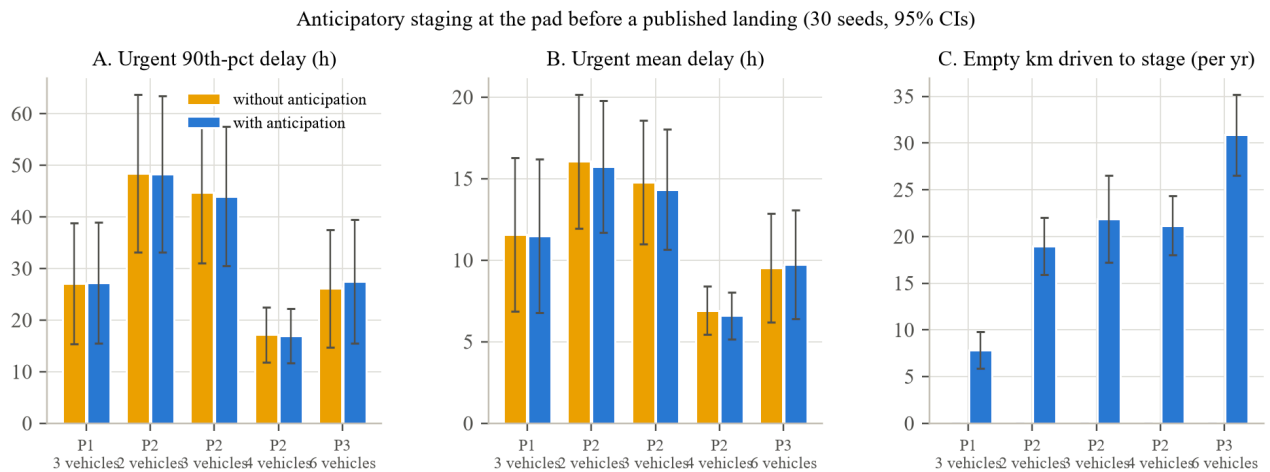


Figure 17. Anticipatory staging before published landings, with and without, on five fleet-and-phase cells (30 seeds; 95% CIs). **A:** urgent 90th-percentile delay. **B:** urgent mean delay. **C:** empty kilometres driven to stage. Differences in A and B are within confidence intervals in every cell.

6.12 The capacity ladder: what capacity moves NASA's manifest

Section 6.4 measured the immovable share for three fleets we chose. Figure 15 removes the fleet from the question. For each phase we drew 300 years of manifests from the cargo classes of Table 2 and NASA's lander cadence, and computed the share of landed mass whose items weigh no more than a given capacity. The curve is a property of the demand model alone and reads directly as a design requirement. In Phase 2, a rover limited to 800 kg (the LTV's full-performance cargo) can move 23% of landed mass; at 1,600 kg (reduced performance), 40%; two LTVs lifting together (3,000 kg), 55%; three (4,500 kg), 64%; a 10-tonne hauler, 88%; a hauler with two LTVs (13,000 kg), 95%; and only a 15-tonne capability reaches 100%. Phase 3 is harder at every rung (16%, 30%, 44%, 57%, 85%, 94%, 100%) because large landers bring a larger share of elements; Phase 1 is easy (63% at 800 kg, 89% at 1,600 kg, 100% at 2,000 kg) because nothing heavier than 2 t lands. The ladder also shows why team lift pays twice: it lifts the LTV-only fleet from 40% to 55–64% of the manifest, and it lifts a single-hauler fleet from 88% to 95%.

Two consequences are worth stating plainly for vehicle designers. First, the 1,600 kg reduced-performance mode of the current LTVs is worth 17 points of Phase-2 manifest over the 800 kg full-performance mode, which is more than any dispatch improvement in this paper. Second, the step from a 10-tonne to a 15-tonne hauler buys the last 12% of the manifest, which is exactly the habitats and power units the base exists to install; a procurement that stops at 10 t will still need a second solution for them.

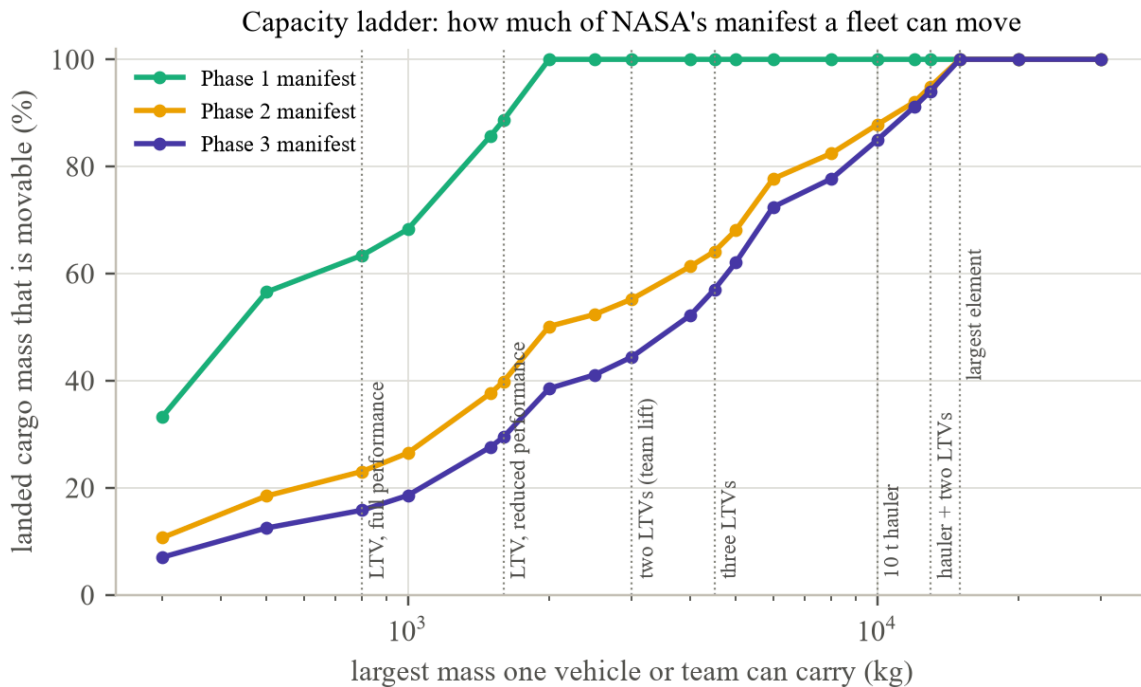


Figure 15. Capacity ladder. Share of landed cargo mass that can be moved by a vehicle or team whose largest payload is the value on the horizontal axis, from 300 manifest-years per phase drawn from Table 2 and NASA's lander cadence. Dotted lines mark existing and proposed capabilities.

6.13 A fragmentation law for in-house fleets

Section 6.3 varied owners with one fleet. Figure 16 varies fleet size, owners and other duties together: fleets of 4, 6 and 8 LTVs, split among 1 to 8 owners, with 0%, 30% or 60% of vehicle time on other duties, 45 in-house cells in all. Plotted against a single quantity, the effective number of vehicles per owner $e = N(1 - d)/k$, the in-

house urgent 90th-percentile delay collapses onto one curve: $25.1 + 20.5/e$ hours fits the 45 cell means with $R^2 = 0.84$, and the scatter around it is of the size of the cells' own confidence intervals. The constant is the delay a single owner with plenty of vehicles still suffers (nights, offload windows, travel); the $1/e$ term is the fragmentation penalty. A fleet split so that each owner has fewer than one effective vehicle pays 20–50 h on the tail; at two effective vehicles per owner the penalty is 10 h; at four it is 5 h. The marketplace's delay in the same fleets is flat at the single-owner value (32–41 h for these all-LTV fleets, which strand their heaviest items and so have a higher floor than the mixed fleets elsewhere in the paper).

The law gives a planner a rule of thumb that the simulation cells alone do not: to know whether pooling will matter at a base, count effective vehicles per owner. NASA's Phase-1 plan, two LTVs owned by two companies that will spend most of their time on NASA crew work, has $e \approx 0.7$ and sits on the steep part of the curve; a single integrator running four rovers full-time has $e = 4$ and would gain little from a marketplace. Panel B shows the same thing as a pooling gain: zero at one owner for every fleet and duty level, rising to 20–38 h at six to eight owners.

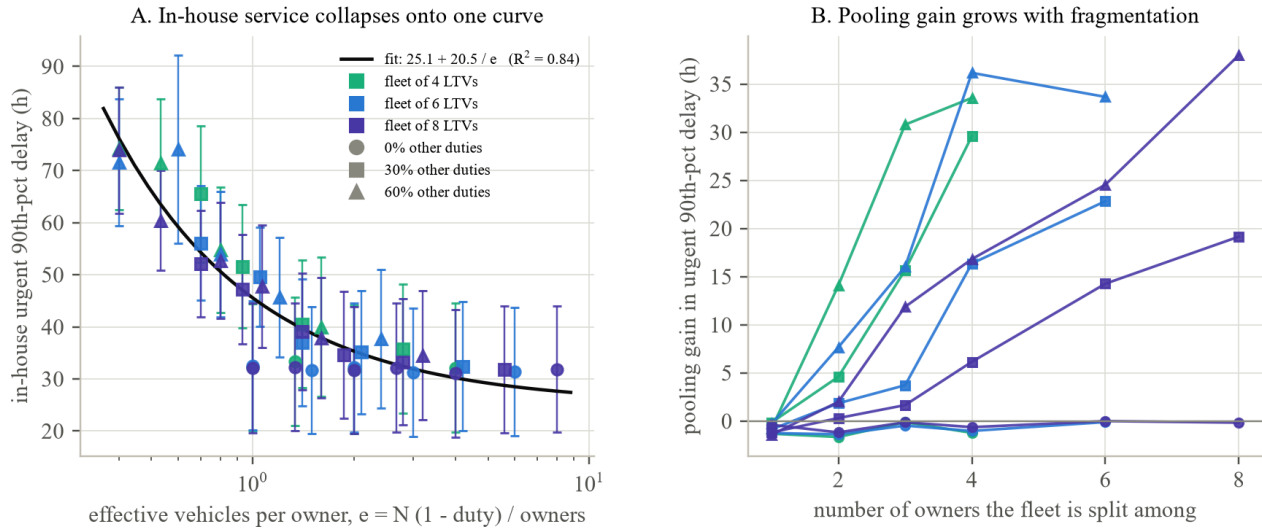


Figure 16. The fragmentation law. **A:** in-house urgent 90th-percentile delay for 45 fleet configurations (fleet size by colour, other-duty share by marker; whiskers are 95% CIs) against effective vehicles per owner, with the fitted curve $25.1 + 20.5/e$. **B:** pooling gain (in-house minus marketplace) against the number of owners, for the same configurations.

6.14 Precision: how many replications settle the answer

Thirty replications rank the mechanisms but cannot pin a mean to a decimal. Figure 18 shows how the 99% confidence interval of the Phase-2 estimates narrows as replications are added, for all four mechanisms, up to the 4,000-replication cap of the first run. The same-day service level, the delivered share and the cost per kilogram settle to within 1% of their means by a few hundred replications. The mean urgent delay needs far more: at 4,000 replications its 99% half-width is still 2.8% in-house and 3.3–3.5% for the pooled rules, while the 90th-percentile delay is at 3.2–4.8%, inside its 5% target. The heavy tail of delays, driven by the few items that wait through a night, is what makes the mean slow to settle. An extended run continued the in-house fleets and pooled greedy cells to the 1% target, which was reached after 32,500 replications, 53,000 replications (half-width 0.99%, 1.00%). The extended run for the batched assignment, marketplace (ours) cells was interrupted by a memory limit on the workstation before it reached that target, so their estimates stand at 4,000, 4,000 replications

with half-widths of 3.3%, 3.5%; the repository script continues them on request. Table 7 lists the settled estimates.

The settled Phase-2 estimates are the ones quoted in Section 6.1: mean urgent delay 14.62 ± 0.15 h in-house, 7.44 ± 0.07 h pooled greedy, 7.54 ± 0.25 h batched assignment and 6.31 ± 0.22 h marketplace; 90th percentiles 38.8 ± 0.4 , 17.6 ± 0.2 , 17.1 ± 0.8 and 15.1 ± 0.7 h; same-day service levels 0.819 ± 0.002 , 0.920 ± 0.001 , 0.924 ± 0.005 and 0.938 ± 0.004 (all 99% CIs). At that precision every pairwise difference among the four rules is distinguishable from zero, including the marketplace's 15% advantage over greedy pooling that the 30-seed cells left in doubt.

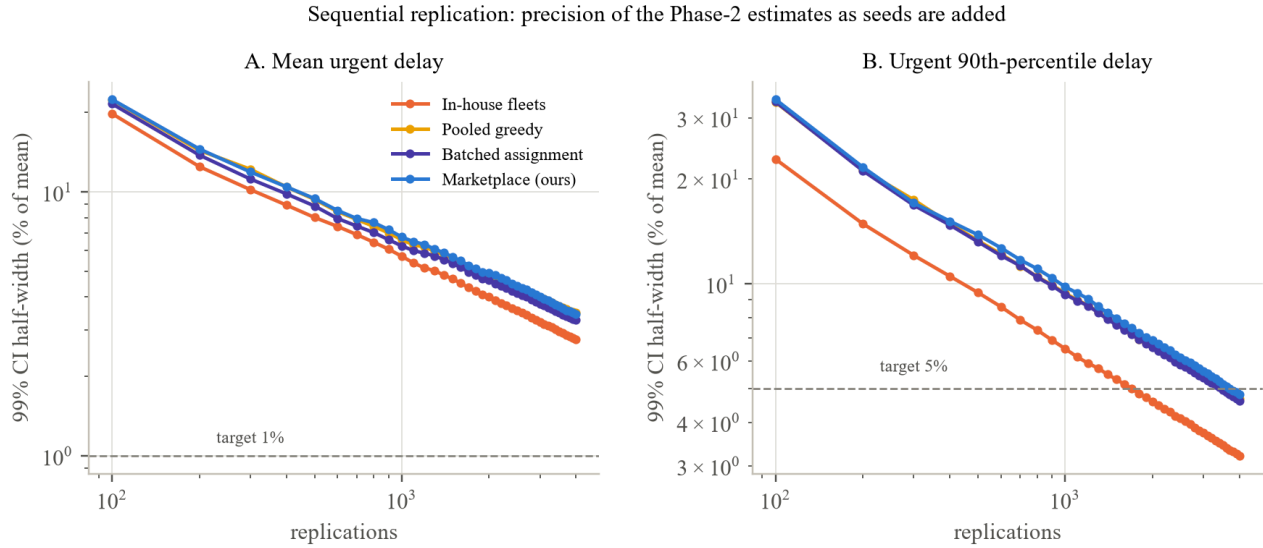


Figure 18. Sequential replication of the Phase-2 default-fleet cells. 99% confidence-interval half-width as a share of the mean, against the number of replications, for the mean urgent delay (A) and its 90th percentile (B); the dashed lines are the targets.

Table 7. Settled Phase-2 estimates from sequential replication (99% confidence intervals). n is the number of replications when the run stopped; a target is reached when the half-width is within the stated share of the mean.

Mechanism	Metric	n	Mean	99% half-width	Half-width / mean	Target	Reached
In-house fleets	Urgent mean delay (h)	32,500	14.62	0.15	0.99%	1%	yes
In-house fleets	Urgent p90 delay (h)	32,500	38.84	0.44	1.13%	5%	yes
In-house fleets	Urgent within 24 h	32,500	0.819	0.002	0.25%	1%	yes
In-house fleets	Delivered share	32,500	0.970	0.000	0.04%	1%	yes
In-house fleets	Cost \$/kg moved	32,500	489.6	0.1	0.03%	1%	yes
Pooled greedy	Urgent mean delay (h)	53,000	7.44	0.07	1.00%	1%	yes
Pooled greedy	Urgent p90 delay (h)	53,000	17.58	0.23	1.29%	5%	yes
Pooled greedy	Urgent within 24 h	53,000	0.920	0.001	0.14%	1%	yes
Pooled greedy	Delivered share	53,000	0.993	0.000	0.01%	1%	yes
Pooled greedy	Cost \$/kg moved	53,000	477.5	0.2	0.04%	1%	yes
Batched assignment	Urgent mean delay (h)	4,000	7.54	0.25	3.27%	1%	no
Batched assignment	Urgent p90 delay (h)	4,000	17.08	0.79	4.61%	5%	yes

Batched assignment	Urgent within 24 h	4,000	0.924	0.005	0.50%	1%	yes
Batched assignment	Delivered share	4,000	0.993	0.001	0.05%	1%	yes
Batched assignment	Cost \$/kg moved	4,000	477.5	0.8	0.16%	1%	yes
Marketplace (ours)	Urgent mean delay (h)	4,000	6.31	0.22	3.46%	1%	no
Marketplace (ours)	Urgent p90 delay (h)	4,000	15.08	0.73	4.82%	5%	yes
Marketplace (ours)	Urgent within 24 h	4,000	0.938	0.004	0.44%	1%	yes
Marketplace (ours)	Delivered share	4,000	0.996	0.000	0.04%	1%	yes
Marketplace (ours)	Cost \$/kg moved	4,000	475.7	0.8	0.18%	1%	yes

7. A reference design: how a lunar logistics marketplace should be built and served

The simulations say what a dispatch layer should do; this section says what the service around it should look like. Figure 19 shows the reference design. It has three layers, a small number of rules that the experiments justify, and a rollout tied to NASA's phases. We write it as a specification a team could build to, not as a vision.

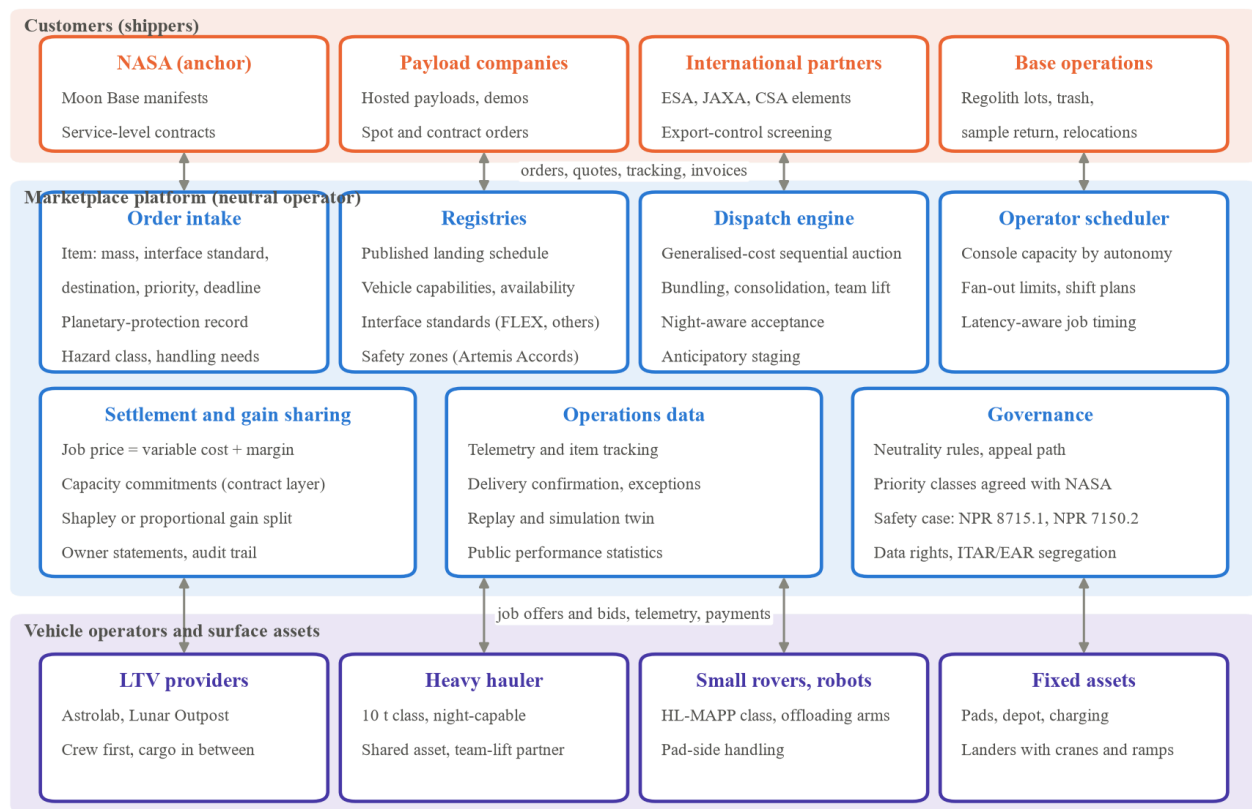


Figure 19. Reference design of the lunar logistics marketplace: customers above, the neutral platform in the middle with its seven functions, vehicle operators and surface assets below. Arrows show the two interfaces the platform owns: orders, quotes, tracking and invoices towards customers; job offers, bids, telemetry and payments towards operators.

7.1 Roles

Customers are whoever owns cargo on the surface: NASA as the anchor customer with service-level contracts, payload companies with hosted demonstrations, international partners with elements of their own, and base

operations itself for regolith, trash, samples and relocations. **Vehicle operators** own and maintain rovers and sell their time; in Phase 2 they are the two LTV providers, the owner of a heavy hauler if one is procured, and the operators of small rovers and pad-side handling robots. The **platform** is a neutral operator that owns no vehicles and no cargo. Neutrality is not decoration: Section 6.3 shows that the value of the service comes entirely from pooling across owners, and an owner-run platform would have both the means and the motive to favour its own fleet.

7.2 The seven functions

Order intake records, for every item, its mass, interface standard, destination, priority class and deadline, its planetary-protection organics record (required for items bound for shadowed sites under NPR 8715.24), its hazard class and handling needs. Items are the unit of the market; manifests are bundles of items. **Registries** hold the published landing schedule, every vehicle's capabilities and availability calendar (including its owner's reserved periods), the interface standards each vehicle accepts, and the current safety zones declared under Section 11 of the Artemis Accords. **The dispatch engine** is Algorithm 1: a generalised-cost sequential auction with bundling, consolidation, team lift and night-aware acceptance, run on every event rather than in batches (Section 6.8), with weights set per base as a policy (Section 6.9). **The operator scheduler** books Earth console time by autonomy level and fan-out, because Section 6.5 shows that a pool of teleoperated rovers is operator-limited before it is vehicle-limited. **Settlement and gain sharing** prices jobs at variable cost plus a margin, holds the contract layer of capacity commitments that Section 6.10 says will carry most of the revenue, and splits pooled gains among owners by a published rule (Shapley or proportional) so that every owner is better off inside the pool than outside it. **Operations data** tracks where every item is, confirms deliveries, logs exceptions and keeps a replayable record; the simulation twin, this testbed with the live manifest and fleet loaded, runs alongside to forecast the week and to test rule changes before they are applied. **Governance** publishes the neutrality rules and the appeal path, agrees priority classes with NASA, holds the safety case against NASA's software and hazard standards, and segregates export-controlled control software from the commercial order layer.

7.3 Rules the experiments justify

- **Pool everything.** Every vehicle not reserved by its owner is offered to every item. This is the whole gain (Sections 6.1–6.3, 6.13).
- **Dispatch on every event; do not batch.** Waiting to thicken the market costs 1–21 h on the urgent tail and buys almost nothing (Section 6.8).
- **Never send a vehicle that cannot get home before dark.** Night awareness removes 1–6 strandings a year and hundreds of wasted operator hours (Sections 6.7, 6.8).
- **Lift together before declaring an item immovable.** Team lift recovers half of what an LTV-only fleet cannot move and most of what a single-hauler fleet cannot (Sections 6.4, 6.12).
- **Hold deferrable bulk for a load, release urgent items at once.** Consolidation to a 70% load or 24 h cuts trips by 4–8% with no cost to urgent cargo (Section 6.8).
- **Price capacity, not jobs.** The fleet is 99% of the cost; contracts for committed capacity should carry the revenue, with the auction allocating work within them (Section 6.10).
- **Measure urgent-cargo delay, not cost.** It is the metric that moves, and it is what the cargo is worth (Section 6.10).
- **Buy or charter one heavy hauler early, and share it.** One shared 10-tonne vehicle takes the immovable share from half of the manifest to a few percent; two take it to zero (Sections 6.4, 6.12).

7.4 Rollout by phase

Phase 1 (through 2029): build the twin, serve by contract. Two LTVs and a handful of landings a year is not a market; $e \approx 0.7$ in the fragmentation law means fragmentation hurts, but there is too little cargo for an auction to matter. The platform's job in this phase is the registries, the order and tracking layer, the operator scheduler and the simulation twin, exercised on Earth freight and on NASA's simulators, and offered to the LTV providers as the system through which they sell their idle time. **Phase 2 (2029–2032): open the pool.** With four to six vehicles and three or more owners the pooling gain is 20–40 h on the urgent tail and the fleet needed for a same-day service level falls from eight to four. This is when the dispatch engine goes live, when a shared hauler should be chartered, and when gain-sharing rules must be in force before the first dispute. **Phase 3 (2032 onward): scale the contract layer.** At 70–150 t a year with elements of 12–15 t, the binding constraints are hauler capacity and operator hours; the platform's leverage shifts to capacity contracts, the night-capable share of the fleet and the autonomy level of the vehicles it dispatches.

7.5 What it would cost to build, and what it could earn

The platform is software plus a small operations team. On the evidence of this project, the dispatch engine, registries, order layer and twin are a few engineer-years; the safety case and export-control segregation are the long poles. We estimate the Phase-1 build at under \$3 million and the Phase-2 operating cost, including a staffed operator-scheduling desk, at \$3–6 million a year. Revenue scales with the fleet capital it coordinates: a platform fee of 3–5% of a \$162–252 million-a-year Phase-2 fleet is \$5–13 million a year, enough to run the service; in Phase 3, with a \$400 million-a-year fleet, it is \$12–20 million. These are estimates, not forecasts, and they depend on an anchor customer choosing to buy surface logistics as a service rather than as vehicles.

8. Discussion

8.1 What the results say to a marketplace operator

The case for a neutral dispatch layer on the Moon rests on three findings, and each comes with a condition. Pooling helps only when the fleet is fragmented, and the market NASA is building is fragmented by design: two LTV providers who may hire out their rovers between NASA missions, several payload companies, international partners, and an agency that has said it will not own the vehicles. The fragmentation law lets an operator check the condition before building: count effective vehicles per owner. The mechanism's value shows up as fleet size: at Phase-2 demand pooled dispatch holds the same service level with four vehicles that in-house contracting cannot reach with eight, and it keeps every owner's heavy items moving instead of leaving the only hauler idle while another company's habitat waits. And the components that carry those results are mundane: pool everybody's vehicles, dispatch on every event, bundle items from the same pad, hold deferrable lots for a day, never send a rover out that cannot get home before dark, and let vehicles lift together. The auction weights buy a further 10–25% on the tail; staging vehicles ahead of landings buys nothing at a base this compact.

Those conditions shape the business. The customer base will be small for a decade, and the anchor customer will be NASA; the Earth analogue is contract trucking with a thin spot layer, not a ride-hailing app [27]. Revenue scales with fleet capital, not with job counts, so a platform fee of a few percent of the fleet's annual cost is the plausible unit (Section 7.5). The platform's technical core is the dispatch rule, the operator scheduler and the data model (manifests, interfaces, organics inventories for shadowed sites, safety-zone geometry under the Artemis Accords [51]); the export-control status of rover command software [52] decides who may run it and from where.

For Roatify specifically, the results support the staged plan we set out before this study: build and run the dispatch platform on Earth freight now, where data and customers exist; test the lunar version in simulation with NASA's published numbers and with rover makers, which is what this paper and its browser testbed do and what NASA's NextSTEP-3 demonstrations require at TRL 6; and launch the surface marketplace when Phase 2 brings enough vehicles and owners for pooling to matter, which the fragmentation law places at fewer than two effective vehicles per owner. The later ambition of building a heavy hauler is also sharpened by the capacity ladder: a 10-tonne vehicle moves 88% of the Phase-2 manifest and a 15-tonne one all of it, and one shared hauler does most of the work of two private ones.

8.2 What the results say to NASA

Four of NASA's stated gaps appear in the results with numbers attached. "Multiple robotic mobility systems working together" recovers about half of the mass an LTV-only fleet cannot move. "High capacity mobility systems" are the only way to move the other half, and the capacity ladder says how much capacity: 10 t for 88% of the Phase-2 manifest, 15 t for all of it. "Autonomy or tele-robotic operation throughout the year" is a factor-of-two to factor-of-three lever on urgent delay and a factor of seven to thirty on operator hours. And a shared dispatch layer is what makes the agency's intention to let LTV providers serve other customers produce a service rather than two private taxis; the fragmentation law says the Phase-1 fleet, at about 0.7 effective vehicles per owner, is exactly where that matters.

8.3 What we would build next

The testbed answers the questions it was built for and leaves three undone. First, an operations demonstration: the dispatch rule running against a higher-fidelity rover simulator with terrain, lighting and a lander manifest for a real Phase-2 mission, which is the TRL 6 artefact NASA's Appendix B asks for. The browser testbed is a step toward it; the next is integration with an existing simulator such as the CARLA-based lunar environment used for NASA's autonomy challenge. Second, a payments and gain-sharing layer: the auction here pays variable cost and reports revenue shares; a real market needs prices that owners will accept and a rule for splitting pooled gains that keeps every owner better off than in-house, for which the collaborative-routing literature offers Shapley-value and proportional rules [9, 10]. Third, strategic behaviour: our vehicles bid their true cost. Whether owners would, and what the mechanism should do if they do not, is an open question that matters more as the market thickens.

9. Limitations

The model is a simplification in ways that could change the numbers, though we think not the ordering of the mechanisms. *Terrain* is a nine-node graph with three slope classes and a fixed tortuosity; real traverse times at a specific site will differ, and we did not model dust, obstacle fields or trafficability by mass, which NASA says make heavy transport "exponentially more difficult" [2]. *Speeds* rest on a 6 km/h autonomous average that is our assumption and a teleoperation table with four published points. *Demand* is Poisson, so it reproduces NASA's annual totals but not the fact that CLPS task orders have slipped by 14 months on average [38]; a real year may have half or twice the landings we draw. *Bulk flows* are our assumption anchored to the IPEX excavation rate; they drive utilisation but not the urgent-cargo results. *Costs* are built from two contract values and a published Starship price; the hauler's cost is a guess, and Section 6.10 shows the cost results scale with it. *Offloading* is a delay, not a resource; cranes, interfaces and lander geometry could create queues we do not see. *Night* is a single block per cycle at a fixed fraction, where real polar sites have several shorter eclipses. *Energy* never bound in

these scenarios and may bind for longer traverses. *Vehicles' other duties* are random blocks; real crew schedules are known in advance and a dispatcher could plan around them. The *mechanism* assumes truthful cost bids and does not model payments, safety-zone deconfliction or communication outages. The *fragmentation law* is fitted on one base layout and one demand model; its constant will differ elsewhere, though we expect the $1/e$ form to hold. Verification shows the engine does what it is specified to do; it does not show that the specification is the Moon. Finally, every result is for one Earth year from an empty base; confidence intervals are wide where landings are few, and several marketplace-versus-greedy differences in the thin Phase-1 setting are not distinguishable from zero at 30 seeds. We report them that way.

10. Broader impact and reproducibility

A dispatch marketplace decides whose cargo moves first and whose vehicle earns. In our runs the generalised-cost bid favours cheap, slow vehicles for deferrable lots and fast ones for urgent items, which is efficient but means that an owner of small, slow rovers earns only low-value work; a real platform will need a visible gain-sharing rule and an appeal path, or small suppliers will not join. Night-aware dispatch protects vehicles at the cost of delaying cargo; who bears that cost should be explicit in contracts. Operating other companies' vehicles raises export-control questions that we flag but do not resolve. On the positive side, pooling spreads revenue more evenly than bilateral contracting (Section 6.1) and keeps heavy items moving for owners who have no hauler of their own.

Everything needed to reproduce this paper is in the repository: the simulator, the scenario definitions, the seed lists, the experiment runner that produced every CSV, the figure scripts that produced every figure and table, the verification test-suite, the parameter notes with sources, and the browser version with its cross-implementation check. Each cell's seeds are 0–29 (0–19 for the latency suite; 0–7 for the weight search; 0 to n for the sequential runs); the demand stream uses the cell seed and the operations stream the seed plus 100,003. Runs are deterministic given a seed. A full rerun of the fourteen suites takes about 40 minutes on a 12-core desktop.

11. Conclusion

We asked whether a lunar logistics marketplace is a real thing or a sentence in a planning document, and built a verified testbed to find out. Four answers hold across 8,000 simulated years. Pooling vehicles across owners roughly halves the delay of urgent cargo and keeps every owner's heavy items moving, but only when the fleet is fragmented: the in-house penalty follows $20/e$ hours in the effective number of vehicles per owner, zero for an integrator and 20–50 h for the fleets NASA is actually buying. A same-day service level at Phase-2 demand needs four pooled vehicles and is out of reach for eight in-house ones, in a market where the fleet is 99% of the cost; a surface move costs \$300 per kilogram moved or about \$5,000 per kilogram of landed cargo, a few percent of what landing it costs. Three limits are beyond any dispatch rule, and the capacity ladder gives the first of them a number: 1.6-tonne rovers move 40% of the Phase-2 manifest, two lifting together 55%, a 10-tonne hauler 88%, and only 15 tonnes of capacity moves all of it; teleoperation from Earth works at lunar latency with a staffed operator pool and collapses at Mars-like latency; and nights, not faults, are the environmental hazard that dispatch must plan around. Among the ideas we tested, batching, anticipatory staging, priority credits and backhaul did not help, and we say so. The reference design in Section 7 turns these findings into a buildable service with a rollout tied to NASA's phases. The next steps are an operations demonstration against a high-fidelity rover simulator, a payments and gain-sharing layer, and a treatment of strategic bidding.

Acknowledgements. The literature search, simulator, experiments, browser testbed and drafting were carried out by the author with Claude (Anthropic) as a research assistant running multiple parallel research agents; all numbers were produced by the released code, all sources were checked against the cited documents, and unverified items are marked in Appendix A.

References

- [1] NASA. *Moon Base Architecture User's Guide*, NP-2026-04-6806-HQ, April 2026. <https://www.nasa.gov/wp-content/uploads/2026/04/moon-base-architecture-users-guide.pdf>
- [2] NASA. *Lunar Mobility Drivers and Needs*, Moon to Mars Architecture Concept Review white paper, 2024. <https://www.nasa.gov/wp-content/uploads/2024/06/acr24-lunar-mobility-drivers-and-needs.pdf>
- [3] NASA. *Lunar Surface Cargo*, Moon to Mars Architecture Concept Review white paper, 2024. <https://www.nasa.gov/wp-content/uploads/2024/06/acr24-lunar-surface-cargo.pdf>
- [4] NASA. “NASA Provides Update on Moon Base Rovers, Landers, Missions,” release 26-046, 26 May 2026. <https://www.nasa.gov/news-release/nasa-provides-update-on-moon-base-rovers-landers-missions/>
- [5] K. Goodliff (NASA ESDMD-SAO). *Lunar Logistics, Mobility, and Cargo*, Industry and Academia workshop slides, February 2025. <https://www.nasa.gov/wp-content/uploads/2025/02/2025-ia-workshop-wp-lunar-logistics-and-mobility.pdf>
- [6] NASA. “NASA Selects Companies to Advance Moon Mobility for Artemis Missions,” 3 April 2024. <https://www.nasa.gov/news-release/nasa-selects-companies-to-advance-moon-mobility-for-artemis-missions/>
- [7] NASA. “NextSTEP-2 Appendix R: Lunar Logistics and Mobility Studies,” awards announced 23–27 January 2025. <https://www.nasa.gov/general/nextstep-r-lunar-logistics-and-mobility-studies/>
- [8] F. Cruijssen, M. Cools, W. Dullaert. “Horizontal cooperation in logistics: opportunities and impediments.” *Transportation Research Part E* 43(2):129–142, 2007.
- [9] M. Gansterer, R. F. Hartl. “Collaborative vehicle routing: a survey.” *European Journal of Operational Research* 268(1):1–12, 2018. <https://arxiv.org/abs/1706.05254>
- [10] M. Frisk, M. Göthe-Lundgren, K. Jörnsten, M. Rönnqvist. “Cost allocation in collaborative forest transportation.” *European Journal of Operational Research* 205(2):448–458, 2010.
- [11] M. Akbarpour, S. Li, S. Oveis Gharan. “Thickness and information in dynamic matching markets.” *Journal of Political Economy* 128(3):783–815, 2020. <https://arxiv.org/abs/1402.3643>
- [12] I. Ashlagi, M. Burq, C. Dutta, P. Jaillet, A. Saberi, C. Sholley. “Edge-weighted online windowed matching.” *Mathematics of Operations Research* 48(2):999–1016, 2023 (ACM EC 2019). <https://www.mit.edu/~jaillet/general/ec19.pdf>
- [13] A. Harris, T. Nguyen. “Long-term relationships and the spot market: evidence from US trucking.” *American Economic Journal: Microeconomics* 17(1):308–353, 2025.
- [14] E. L. Gralla, S. Shull, O. L. de Weck. “A modeling framework for interplanetary supply chains.” AIAA SPACE 2006; and S. A. Shull, *Integrated Modeling and Simulation of Lunar Exploration Campaign Logistics*, S.M. thesis, MIT, 2007. <https://dspace.mit.edu/handle/1721.1/39711>
- [15] T. Ishimatsu, O. L. de Weck, J. A. Hoffman, Y. Ohkami, R. Shishko. “Generalized multicommodity network flow model for the Earth–Moon–Mars logistics system.” *Journal of Spacecraft and Rockets* 53(1):25–38, 2016. <https://dspace.mit.edu/handle/1721.1/110236>
- [16] K. Ho. *Dynamic Network Modeling for Spaceflight Logistics with Time-Expanded Networks*, Ph.D. thesis, MIT, 2015; B. B. Jagannatha, K. Ho, “Event-driven network model for space mission optimization,” *Journal of Spacecraft and Rockets* 57(3), 2020. <https://arxiv.org/abs/1904.09364>
- [17] M. Shirley, E. Balaban. *An Overview of Mission Planning for the VIPER Rover*, NASA, 2 June 2022. https://www.nasa.gov/wp-content/uploads/2022/05/overview_of_mission_planning_for_the_viper_rover.pdf
- [18] A. Krawciw, L. Antonyshyn, S. Lilge, N. Olmedo, F. Rehmatullah, M. Desjardins-Goulet, P. Toupin, T. D. Barfoot. “Sharing the load: autonomous multi-rover cargo transport.” arXiv:2510.18766, 2025. <https://arxiv.org/abs/2510.18766>
- [19] E. Tuci, M. H. M. Alkilabi, O. Akanyeti. “Cooperative object transport in multi-robot systems: a review of the state-of-the-art.” *Frontiers in Robotics and AI* 5:59, 2018; H. Farivarnejad, S. Berman, “Multirobot control strategies for collective transport,”

- [20] M. B. Dias, R. Zlot, N. Kalra, A. Stentz. “Market-based multirobot coordination: a survey and analysis.” *Proceedings of the IEEE* 94(7):1257–1270, 2006.
- [21] B. P. Gerkey, M. J. Mataric. “A formal analysis and taxonomy of task allocation in multi-robot systems.” *International Journal of Robotics Research* 23(9):939–954, 2004.
- [22] S. Koenig, C. Tovey, M. Lagoudakis, V. Markakis, D. Kempe, P. Keskinocak, A. Kleywegt, A. Meyerson, S. Jain. “The power of sequential single-item auctions for agent coordination.” AAAI 2006, pp. 1625–1629. <https://cdn.aaai.org/AAAI/2006/AAAI06-266.pdf>
- [23] M. G. Lagoudakis et al. “Auction-based multi-robot routing.” *Robotics: Science and Systems I*, 2005. <https://roboticsproceedings.org/rss01/p45.html>
- [24] H.-L. Choi, L. Brunet, J. P. How. “Consensus-based decentralized auctions for robust task allocation.” *IEEE Transactions on Robotics* 25(4):912–926, 2009.
- [25] C. Caplice, Y. Sheffi. “Optimization-based procurement for transportation services.” *Journal of Business Logistics* 24(2), 2003; “Combinatorial auctions for truckload transportation,” in *Combinatorial Auctions*, MIT Press, 2006.
- [26] R. Chen, S. Kim, H. Wang, X. Wang. “Posted price versus hybrid mechanisms in freight transportation marketplaces.” arXiv:2106.00923, 2021.
- [27] DAT Freight & Analytics. “Dry van report: the contract freight market continues to soften” (about 15% of loads on the spot market); ATRI, *An Analysis of the Operational Costs of Trucking*, 2024 (deadhead 16–21%). <https://www.dat.com/blog/dry-van-report-the-contract-freight-market-continues-to-soften>
- [28] NASA NTRS 20250000703, Artemis communications latency (up to about 10 s one way including processing). <https://ntrs.nasa.gov/citations/20250000703>
- [29] H. L. Litaker, Z. Q. Li, K. H. Beaton, J. F. Lewis. *Lunar Terrain Vehicle (LTV) Remote Teleoperation Studies Under Four Lunar Communication Latencies*, NASA TM-20240001217, 2024; follow-up NTRS 20240011392, 2025. <https://ntrs.nasa.gov/citations/20240001217>
- [30] M. Seo, S. Gupta, Y. Ham. “Evaluation of performance and mental workload during time delayed teleoperation for the lunar surface construction.” <https://par.nsf.gov/servlets/purl/10479687>
- [31] W. R. Ferrell. *Remote Manipulation with Transmission Delay*, MIT, 1965; T. B. Sheridan, “Space teleoperation through time delay: review and prognosis,” *IEEE Transactions on Robotics and Automation* 9(5):592–606, 1993.
- [32] D. R. Olsen, S. B. Wood. “Fan-out: measuring human control of multiple robots.” CHI 2004. <https://hci.rwth-aachen.de/materials/conferences/CHI2004/1p231.pdf>
- [33] Komatsu, FrontRunner autonomous haulage system (one controller per fleet); Equipment World explainer (up to about 30 trucks per operator). <https://www.komatsu.com/en-au/technology/smart-solutions/smart-mining/autonomous-haulage-system>
- [34] V. Grimm et al. “The ODD protocol for describing agent-based and other simulation models: a second update to improve clarity, replication, and structural realism.” *JASSS* 23(2):7, 2020. <https://www.jasss.org/23/2/7.html>
- [35] T. Monks, C. S. M. Currie, B. S. Onggo, S. Robinson, M. Kunc, S. J. E. Taylor. “Strengthening the reporting of empirical simulation studies: introducing the STRESS guidelines.” *Journal of Simulation* 13(1):55–67, 2019.
- [36] NASA. “IPEX TRL 5 Design Overview,” ASCEND 2024 (10,000 kg of regolith per day). https://ntrs.nasa.gov/api/citations/20240008162/downloads/ASCEND24_IPEX%20TRL%205%20Design%20Overview.pdf
- [37] NASA. *Lunar Logistics Drivers and Needs*, 2023 Architecture Concept Review white paper (packaging overhead 75–150%). <https://www.nasa.gov/wp-content/uploads/2024/01/lunar-logistics-drivers-and-needs.pdf>
- [38] NASA. Commercial Lunar Payload Services reference page and task-order table. <https://www.nasa.gov/reference/commercial-lunar-payload-services/> ; NASA OIG, IG-24-013, June 2024.
- [39] LROC. “Islands in the Dark” and Shackleton rim illumination maps. <https://www.lroc.asu.edu/images/1105>
- [40] Rio Tinto autonomous haulage performance (about 1,000 additional operating hours per truck per year; 15% lower load-and-haul unit cost), *Mining Journal*; *Mining* 6(3):61, MDPI (availability about 88%, utilisation about 95%). <https://doi.org/10.3390/mining6030061>

- [41] NASA NSSDC. Apollo Lunar Roving Vehicle (210 kg, 490 kg payload, 2×36 V 121 Ah batteries, 92 km design range). https://nssdc.gsfc.nasa.gov/planetary/lunar/apollo_lrv.html
- [42] H. W. Kuhn. “The Hungarian method for the assignment problem.” *Naval Research Logistics Quarterly* 2(1–2):83–97, 1955.
- [43] A. M. Law. “Statistical analysis of simulation output data: the practical state of the art.” *Proceedings of the Winter Simulation Conference*, 2004 and 2023; A. M. Law, *Simulation Modeling and Analysis*, 5th ed., McGraw-Hill, 2015. <https://www.informs-sim.org/wsc23papers/124.pdf>
- [44] J. C. Castillo, D. Knoepfle, E. G. Weyl. “Surge pricing solves the wild goose chase.” ACM EC 2017.
- [45] B. J. Mellinkoff, W. Bailey, M. M. Spydell, J. O. Burns. “Quantifying operational constraints of low-latency telerobotics for planetary surface operations.” 2018. <https://arxiv.org/html/1710.01254>
- [46] SpaceX lunar cargo price, \$100 million per metric ton, as published on [spacex.com](https://www.spacex.com) and reported by Universe Magazine, “Lunar logistics moves to a shared flight model.” <https://universemagazine.com/en/lunar-logistics-moves-to-a-shared-flight-model/>
- [47] SpacePolicyOnline. “Firefly lands on the Moon” (Blue Ghost Mission 1: \$101.5 million for ten NASA payloads totalling about 94 kg). <https://spacepolicyonline.com/news/firefly-lands-on-the-moon/>
- [48] PwC. *Building the Lunar Economy: Sectorial Forecasts and Market Opportunity*, January 2026 (US\$93.9–127.3 billion cumulative revenue to 2050). Summary: <https://spaceq.ca/pwc-report-lunar-economy-projected-to-generate-up-to-us127-billion-by-2050/>
- [49] Deloitte. *Building the Lunar Economy*, 28 August 2026 (US\$343–566 billion cumulative value to 2050; transportation US\$206 billion). Summary: <https://fortune.com/2026/08/28/deloitte-sees-a-566-billion-lunar-economy-by-2050/>
- [50] T. J. Colvin, K. W. Crane, R. Lindbergh, B. Lal. *Demand Drivers of the Lunar and Cislunar Economy*, IDA Science and Technology Policy Institute, D-13219, 2020. <https://ida.org/research-and-publications/publications/all/d/de/demand-drivers-of-the-lunar-and-cislunar-economy>
- [51] The Artemis Accords, Section 11 (deconfliction and safety zones), 13 October 2020; 76 signatories as of 25 September 2026. <https://www.nasa.gov/wp-content/uploads/2022/11/Artemis-Accords-signed-13Oct2020.pdf>
- [52] Holland & Knight. “New export control rules present key regulatory changes for space-related items, activities,” November 2024 (ECCN 9A515/9D515 for planetary rovers and software). <https://www.hkclaw.com/en/insights/publications/2024/11/new-export-control-rules-present-key-regulatory-changes-for-space>
- [53] S. B. Stancliff, J. M. Dolan, A. Trebi-Ollennu. “Planning to fail: reliability as a design parameter for planetary rover missions.” PerMIS 2007.
- [54] NASA. “Moon Base Systems: Achieving the Vision,” 11 June 2026 (Phase-3 logistics target of up to eight metric tons per 28-day mission). <https://www.nasa.gov/moonbase-systems/>
- [55] Astrolab. FLEX Services and the FLEX Universal Payload interface (open format, up to 1,600 kg, >3 m³). <https://astrolab.space/flex-services>
- [56] Y. Zhang et al. “Research on task allocation method for multi-agent systems on the Moon with a distributed architecture.” IAC-24, A3.IP.160, 2024.
- [57] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, S. Yao. “Reflexion: language agents with verbal reinforcement learning.” arXiv:2303.11366, 2023; A. Gandhi et al., “Recursive agent optimization,” arXiv:2605.06639, 2026; J. Pineau et al., “Improving reproducibility in machine learning research,” *JMLR* 22, 2021. (Methodological models for the experimental protocol: baselines, ablations, train/test generalisation, reproducibility checklist.)
- [58] S. Ichoua, M. Gendreau, J.-Y. Potvin. “Exploiting knowledge about future demands for real-time vehicle dispatching.” *Transportation Science* 40(2):211–225, 2006; S. Mitrović-Minić, G. Laporte, “Waiting strategies for the dynamic pickup and delivery problem with time windows,” *Transportation Research Part B* 38(7):635–655, 2004.
- [59] K. Hoad, S. Robinson, R. Davies. “Automated selection of the number of replications for a discrete-event simulation.” *Journal of the Operational Research Society* 61(11):1632–1644, 2010. <https://wrap.warwick.ac.uk/5071/>

Appendix A. Parameters, sources and confidence

Confidence: **high** = read in the primary document; **med** = reputable secondary report or primary seen only in summary; **low** = derived or single-source; **assumed** = our choice, swept in Section 6.10 where it matters.

Parameter	Value used	Basis	Confidence
Phase landed mass	~4,000 / ~60,000 / ~150,000 kg; 21 / 24 / 28 landings	Moon Base guide p.3 [1]	high
Recurring logistics	2,000–6,000 kg per crewed mission; 2,500–10,000 kg/yr	[2, 3]	high
Habitation-class elements	12,000–15,000 kg	[2, 3]	high
Pad standoff; farthest site	>1,000 m; up to 5,000 m	[2]	high
Slopes	up to 20°; >10° common	[2]	high
Tortuosity factor	1.3	our choice	assumed
LTV cargo capacity	800 kg full / 1,600 kg reduced performance; 1,500 kg used	[2, 5]; Astrolab 1,500–1,600 kg [55]	high / med
LTV speed	10 km/h design target; 6 km/h autonomous average used	[1]; Astrolab CLV-1 >6 mph, Pegasus >9 mph [4]	med / assumed
LTV cost	\$219–220M task order + ~\$234M MK1 delivery = \$450M	[4]	high / derived
Hauler	10,000 kg, 4 km/h, \$600M, night-capable	“high capacity mobility” gap [1, 5]; Starship \$100k/kg [46]	assumed
Small rover	300 kg, 3 km/h, \$60M	Lunar Outpost HL-MAPP 250 kg / 200 kg / 3 m/s	med / assumed
Lander payloads	CLPS 100–500 kg; MK1 2,000–3,000 kg; large 12,000–30,000 kg	CLPS task orders 0.07–0.475 t [5]; MK1 3 t; MK2 20–30 t	high / med
Offload delay	6–24 h (CLPS), 12–36 h (MK1), 24–72 h (large)	our choice	assumed
Dark share of cycle	0.15 default (about 106 h)	Shackleton rim 90–94% lit [39]; 120 h survival target [1]	med
Night speed factor (night-capable)	0.6	our choice	assumed
Light-time RTT; processing	2.6 s; +2.4 s default	[28, 45]	high / med
Teleop speed model	table 3.24 / 2.56 / 2.03 / 1.76 km/h at 0 / 4 / 6 / 8 s, interpolated; move-and-wait extrapolation beyond 8 s	[29]; [31]	med (numbers from summary)
Teleop handling factor	1 + 0.25·RTT	139% at 1.5 s, 174% at 3 s [30]	med
Fan-out	supervised 4; autonomous 20	Olsen & Wood 1–9 [32]; mining up to 30 [33]	med

MTBF / MTTR	500/24 h (LTV), 400/36 h (hauler), 300/12 h (small)	chosen to give 92–96% availability as in autonomous haulage [40]; no rover MTBF published [53]	assumed
Other-duty share	0.3 in 48 h blocks	LTVs serve NASA first, hired out between missions [6]; a trade estimate of 25% commercial use	low
Energy	350 / 900 / 120 Wh/km by class	LRV-derived ~95 Wh/km at 700 kg [41], scaled	low
Operator cost	\$1,000 per console hour	our choice; swept \$200–\$5,000	assumed
Value of time by priority	\$5,000 / \$1,000 / \$100 per hour	our choice	assumed
Wear share ω	0.5	our choice	assumed
Return flow	30% of items; 20–60% of mass; 5–30 days later	trash and sample return are Appendix R topics [7]	assumed
Bulk flows	P1 30 t/yr; P2 500 t/yr; P3 1,500 t/yr	IPEX 10 t/day [36]; swept 0–3,000	assumed
Anticipation window and margin	48 h; arrive 1 h before first item	our choice	assumed
Landed cost per kg	CLPS ~\$1.08M (Blue Ghost M1); MK1 ~\$78k if full; Starship \$100k published	[47]; [4]; [46]	high / derived / med

Table A2. Mission manifest behind Figure 3A (base case, tonnes of surface payload at or near the south pole). Non-polar CLPS flights are excluded. Dates reflect NASA's May–June 2026 statements and the New Glenn loss of 28 May 2026.

Year	Low	Base	High	Base-case content
2026	0.0	0.5	0.7	Griffin-1 with the FLIP rover (~0.5 t)
2027	0.2	0.8	3.5	Blue Moon MK1 Pathfinder, IM-4, Draper CP-12, VIPER on a second MK1
2028	1.0	4.5	12	Two LTV deliveries on MK1 with co-manifested cargo; Artemis IV crew logistics; MoonFall drones
2029	1.5	6	15	Recurring logistics begins (NASA: 2,500–10,000 kg/yr); Eagle rover on Starship; IM-5
2030	3	20	35	Fission Surface Power unit (≤ 15 t) plus recurring logistics
2031	3	12	30	Recurring logistics; ESA Argonaut (1.5 t); ISRU pilot elements
2032	5	25	45	JAXA pressurised rover (12–15 t class); recurring logistics up to 8 t per mission

Appendix B. Verification table

Each check compares a simulator output with an independent reference and passes when the relative error is within its tolerance. The same thirteen checks run in the Python test-suite (`tests/test_verification.py`) and, with the M/G/1 check replaced by a determinism check, on the Verification tab of the browser version.

Table B1. Verification checks of the Python implementation (all pass).

Check	Reference	Simulated	Rel. error	Tolerance	Basis	Result
Travel time on a flat two-edge path	0.37442 h	0.37442 h	0.000%	0.0%	distance / speed; path PAD_A-DEPOT-BASE, both edges flat	PASS

Steep-edge travel time doubles	0.67827 h	0.67827 h	0.000%	0.0%	slope class 'steep' halves speed	PASS
Teleoperated speed at 0 s RTT	3.24 km/h	3.24 km/h	0.000%	1.0%	NASA JSC LTV teleoperation trial average speed	PASS
Teleoperated speed at 4 s RTT	2.56 km/h	2.56 km/h	0.000%	1.0%	NASA JSC LTV teleoperation trial average speed	PASS
Teleoperated speed at 6 s RTT	2.03 km/h	2.03 km/h	0.000%	1.0%	NASA JSC LTV teleoperation trial average speed	PASS
Teleoperated speed at 8 s RTT	1.76 km/h	1.76 km/h	0.000%	1.0%	NASA JSC LTV teleoperation trial average speed	PASS
Dark share of time	0.15 fraction	0.15 fraction	0.000%	0.5%	night_fraction parameter; sampled over 10 cycles	PASS
Mass conservation (marketplace)	3.3449e+05 kg	3.3449e+05 kg	0.000%	0.0%	ready mass = delivered + stranded + backlog at horizon	PASS
Mass conservation (pooled_greedy)	3.3449e+05 kg	3.3449e+05 kg	0.000%	0.0%	ready mass = delivered + stranded + backlog at horizon	PASS
Items above team capacity (2 x 1,500 kg) never delivered	0 items	0 items	0.000%	0.0%	13 items heavier than 3,000 kg in six large landings	PASS
Same landed manifest for every mechanism (seed 11)	1 bool	1 bool	0.000%	0.0%	demand drawn from a stream separate from operations	PASS
Annual fleet amortisation	117 \$M	117 \$M	0.000%	0.0%	sum of unit cost / calendar life over the fleet	PASS
M/G/1 mean queue wait (Pollaczek-Khinchine)	0.21934 h	0.21823 h	0.507%	5.0%	rho=0.190, lambda=2.47/day, E[S]=1.85 h, n=18117 items over 20 years	PASS

Appendix C. STRESS-DES reporting checklist

Item	Where answered
1.1 Purpose of the model	Section 3.1
1.2 Model outputs	Section 3.6, Section 5 (metrics)
1.3 Experimentation aims	Section 1 (RQ1–RQ5), Table 4
2.1 Base model overview diagram	Figure 2
2.2 Base model logic	Sections 3.2–3.6
2.3 Scenario logic	Table 4; experiments/run_all.py
2.4 Algorithms	Section 4, Algorithm 1; lunarsim/dispatch.py
2.5 Components (entities, activities, resources, queues)	Section 3.1, 3.6
3.1 Data sources	Appendix A, References
3.2 Pre-processing	Section 3.4 (latency table)
3.3 Input parameters	Tables 2, 3, A1

3.4 Assumptions	Appendix A (“assumed” rows), Section 9
4.1 Initialisation and warm-up	Section 5 (empty base, no warm-up deletion; justification given)
4.2 Run length	One Earth year (8,760 h), event-driven
4.3 Estimation approach	30 (or 20) replications; 95% CIs; common random numbers; paired differences; sequential replication to 99% / 1% for headline cells (Section 5.3, 6.14)
5.1 Software	Python 3.12.3, NumPy, pandas, Matplotlib (SciPy optional); JavaScript port; Windows 11
5.2 Random sampling	NumPy PCG64 (Python), sfc32 (browser); demand stream seeded with the cell seed, operations stream with seed + 100,003
5.3 Model execution	Event heap ordered by (time, sequence number); batched mechanisms act at ticks only
5.4 System specification	12-core desktop; 0.1–3 s per simulated year; ~40 min for the fourteen suites
6.1 Code access	<code>lunar-logistics-research/</code> (simulator, scenarios, seeds, tests, figure scripts, browser version)

Appendix D. Implementation notes on the mechanisms

Feasibility: a vehicle can carry an item if the item's interface standard is in the vehicle's accepted set and the item's mass does not exceed the vehicle's capacity. Teams: for an item heavier than every free vehicle, candidate vehicles are those that are free, team-capable and compatible, sorted by capacity; the first pair or triple whose capacities sum to at least the item's mass is chosen. Routing within a bundle: all permutations of up to four distinct destinations are evaluated on slope-weighted distance from the pad. Insertion test: an item joins a bundle if its marginal duration is below 80% of the duration of a separate single-item job for the same vehicle. Pool cap: when more than 40 items are waiting, one auction call considers the 40 most urgent and oldest; remaining items are considered at the next event. Anticipation: a landing within 48 h whose pad has no present or inbound vehicle triggers a staging move by the nearest unused free vehicle, timed to arrive one hour before the earliest possible item release, unless the move would be caught by the night. Payments: each job is paid its variable cost (Section 3.5) and credited to the vehicle owners in equal shares for teams. Batched assignment uses a minimum-cost assignment (SciPy's `linear_sum_assignment` when available, otherwise a shortest-augmenting-path Hungarian implementation verified against SciPy on 300 random matrices) with cost = single-item job duration – $0.05 \times$ waiting age; infeasible pairs cost 10^6 . In-house carrier assignment: owners without vehicles are mapped round-robin to carriers in sorted order at the start of the run.